Lesson 1: Introduction to Business Process Management

Why You Should Learn BPM, Can Become a

- **Business Process Analyst**
  Studies how work is done and finds better ways to do it.
- **BPM Developer**
  Builds and sets up BPM systems using software tools.
- **IT Business Analyst**
  Connects business needs with IT solutions.
- **Process Automation Specialist**
  Automates repetitive tasks using tools like RPA.
- **BPM Consultant**
  Advises organizations on how to use BPM to improve work.

What is Business Process Management (BPM)?

Simple Meaning:
BPM is how a company manages and improves the way work is done so everything becomes faster, clearer, and more efficient.

Simplified Definitions

1. Workflow Management Coalition / BPM.com
BPM means modeling, automating, running, controlling, and improving business activities to achieve company goals.

Example: A bank uses BPM to automate the loan approval steps, data entry, checking documents, and sending approval messages.

2. Gartner
BPM is about managing processes (not single tasks) to improve performance. Processes connect people, information, systems, and tools to deliver value to customers.

Example: Delivering an online order involves website, payment system, warehouse, courier—BPM connects all of these smoothly.

General Explanation
BPM is a mix of art and science that helps a company improve how its processes work.
Example: A hospital uses BPM to reduce patient waiting time by redesigning registration, doctor assignment, and payment steps.

Another simple explanation

BPM is managing and improving the steps of work inside an organization to get better results.

Example: A restaurant uses BPM to speed up food orders by improving the steps from taking orders → cooking → serving.

What BPM *is*

- **A way to manage processes**
  Example: A hotel uses BPM to manage the full check-in process from booking to room assignment.
- **A way to identify all processes inside a company**
  Example: A supermarket maps out billing, stocking, and delivery processes.
- **A business-led method or project**
  Example: Management decides how to improve sales or customer service processes.
- **A base for continuous improvement**
  Example: A factory uses BPM to keep improving its production line over time.

What BPM is *NOT*

- **Not only an improvement method** (but it helps improvement)
  Example: Six Sigma is an improvement tool, BPM is broader.
- **Not an IT-only solution**
  Example: Software supports BPM, but business teams decide the processes.
- **A tactical level tool** (it is strategic)
  Example: BPM helps decide big steps like redesigning company-wide workflows.

Why BPM is needed

- **Improve efficiency**
  Example: A bank reduces loan approval time from 5 days to 1 day.
- **Enhance customer satisfaction**
  Example: Faster deliveries make online shoppers happier.
- **Support continuous improvement**
  Example: A call center keeps updating its scripts and workflow.
- **Increase agility**
  Example: A company quickly changes its process when a new law comes.
- **Reduce costs**
  Example: Automating invoices reduces manual work.
- **Improve compliance & risk management**
  Example: A hospital follows proper patient data rules.
- **Enable better decision-making**
  Example: Managers see process data and find what's slowing down work.
- **Drive innovation**
  Example: A company adds online self-service after reviewing its processes.

- **Empower employees**
  Example: Staff feel confident when processes are clear and easy to follow.

Key Components – Processes Everywhere

- **Processes exist in every organization**
  Example: A school has admission, grading, and fee collection processes; a bank has loan processing and account opening processes.
- **What a process is**
  A **process** is a series of connected tasks or activities done to achieve a specific goal.
  Tasks can be done by people or automated systems.
  Example: Online shopping process: customer places order → system checks stock → payment processed → order shipped.

- **Ingredients of a business process**

  - **Events and activities:** Things that happen or actions taken.
    Example: Receiving an order, packing items.
  - **Decision points:** Moments where choices are made.
    Example: Approve or reject a loan application.
  - **Actors:** People or systems performing the tasks.
    Example: Cashier, warehouse robot.
  - **Physical objects:** Tangible items used or produced.
    Example: Products in a warehouse.
  - **Information objects:** Data or documents used in the process.
    Example: Invoices, customer details.
  - **Final outcome:** The result of the process.
    Example: Customer receives their order.
  - **Customer:** The end recipient of the process output.
    Example: Shopper, client, citizen.

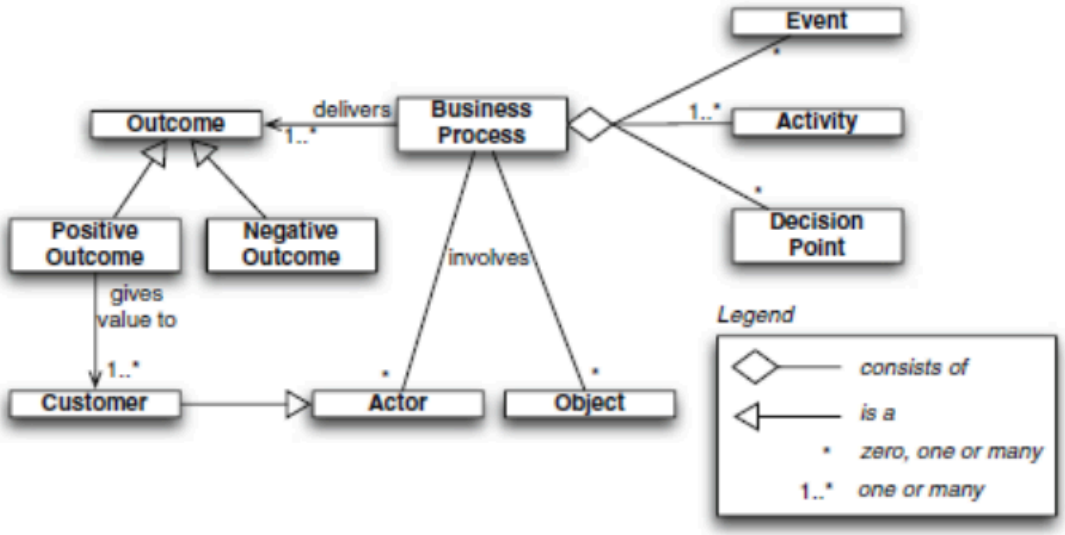
Ingredients of a Business Process

- **Events**
  - Events are things that happen instantly and trigger, change, or end a process.
  - Can be internal (inside organization) or external (outside).
  - *Example:* A delivery truck arrives at a construction site, starting the equipment inspection process.
- **Activities**
  - Activities are tasks or actions done in the process, which take time. Can be manual, automated, or both.
  - *Example:* Site engineer inspects the delivered equipment.
  - Simple activities are called **tasks**.
- **Decision Points**
  - Points where choices determine the next step in the process.

- *Example:* After inspection, engineer decides to accept or return the equipment. This affects the following steps.

*A process also involves

- **Actors**
  - People, organizations, or software performing tasks in the process.
  - *Example:* Warehouse staff, automated inventory system.
- **Physical Objects**
  - Tangible items used or produced in the process.
  - *Example:* Equipment, raw materials, documents.
- **Informational Objects**
  - Data or documents used in the process.
  - *Example:* Digital invoices, inspection reports.
- **Final Outcome**
  - Result(s) of the process, ideally delivering value to participants.
  - Positive outcome = value delivered, Negative outcome = value not fully achieved.
  - **Customer:** The one who receives or benefits from the process output.
  - *Example:* Completed inspection report delivered to project manager, ensuring safe use of equipment.



Exercise 02: Identify the event, activities, decision points, actors, physical objects, Information Objects and final outcome from the following scenario.

Online Food Delivery of a restaurant

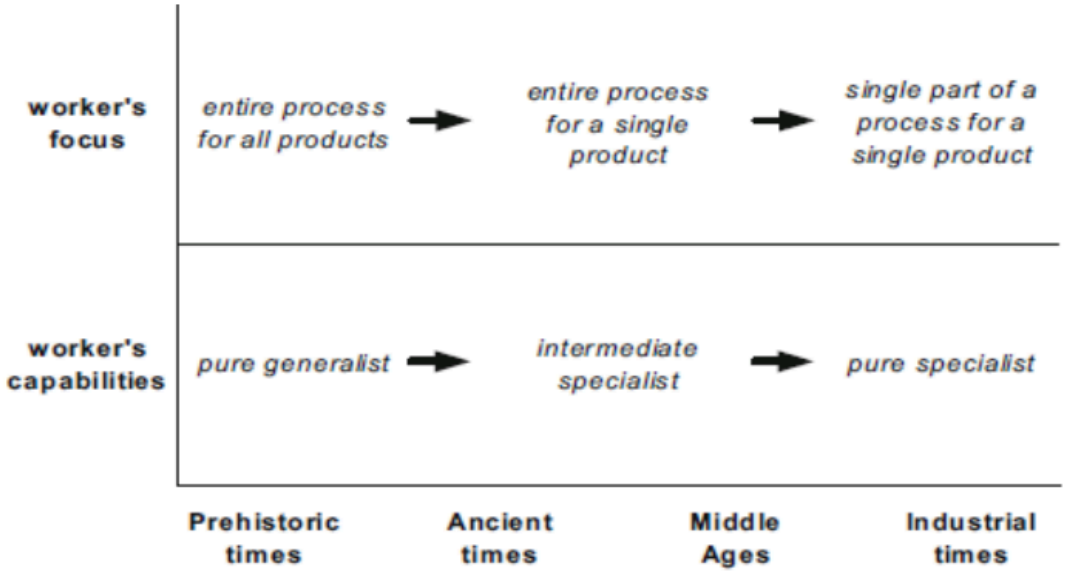
- In this scenario, the process begins when a customer opens a food delivery mobile application and places an order from a chosen restaurant. Once the order is submitted, the restaurant receives and confirms it before the kitchen staff begins preparing the meal. After the food is ready, a delivery rider is assigned through the app to collect and deliver the order to the customer's location. During this process, several decision points are encountered, such as verifying whether the payment has been successfully processed, confirming the restaurant's acceptance of the order, and checking the availability of a delivery rider. The key actors involved include the customer, restaurant staff, delivery rider, and the food delivery application system. The main physical objects handled in the process are the food, delivery box, and delivery bike, while essential information objects include the order details, payment receipt, delivery tracking information, and customer feedback data. The process concludes when the customer receives the meal in good condition and provides feedback, marking the successful completion of the transaction.

Online Food Delivery Process – Business Process Ingredients

- **Event:**
  - The customer opens the food delivery app and places an order.
- **Activities:**
  - The restaurant confirms the order.
  - The kitchen staff prepares the meal.
  - Delivery rider collects and delivers the order.
  - The customer receives the meal.
- **Decision Points:**
  - Has the payment been successfully processed?
  - Does the restaurant accept the order?
  - Is a delivery rider available?
- **Actors:**
  - Customer
  - Restaurant staff

    - Kitchen staff
    - Delivery rider
    - Food delivery app system
- **Physical Objects:**
    - Food
    - Delivery box
    - Delivery bike
- **Information Objects:**
    - Order details
    - Payment receipt
    - Delivery tracking information
    - Customer feedback
- **Final Outcome:**
    - The customer receives the meal in good condition and gives feedback, completing the transaction successfully.

History of BPM



Historical Milestones in BPM

- **BPR (Business Process Reengineering):**
  - A major event was Ford investing heavily in Mazda in the 1980s, which drove new process approaches.
- **Technology:**
  - New IT systems emerged to support processes, including **ERP systems** and **Workflow Management Systems (WfMSs)**.

Aspect	BPR (Business Process Reengineering)	BPM (Business Process Management)
Meaning	Completely redesigning a business process from the beginning.	Managing and improving existing business processes continuously.
Main Idea	**Start from scratch** and build a new process.	**Improve step by step** the current process.
Level of Change	Very big change (radical change).	Small or gradual improvements.
Frequency	Done rarely (only when major change is needed).	Done continuously to improve performance.
Risk	High risk because everything changes.	Low risk because changes are small.
Goal	Achieve dramatic improvement in cost, speed, or service.	Make processes more efficient and organized over time.
Example (Hotel)	A hotel removes manual booking completely and builds a new **online reservation system.**	A hotel keeps the same reservation system but improves it by **reducing booking steps or adding faster check-in.**

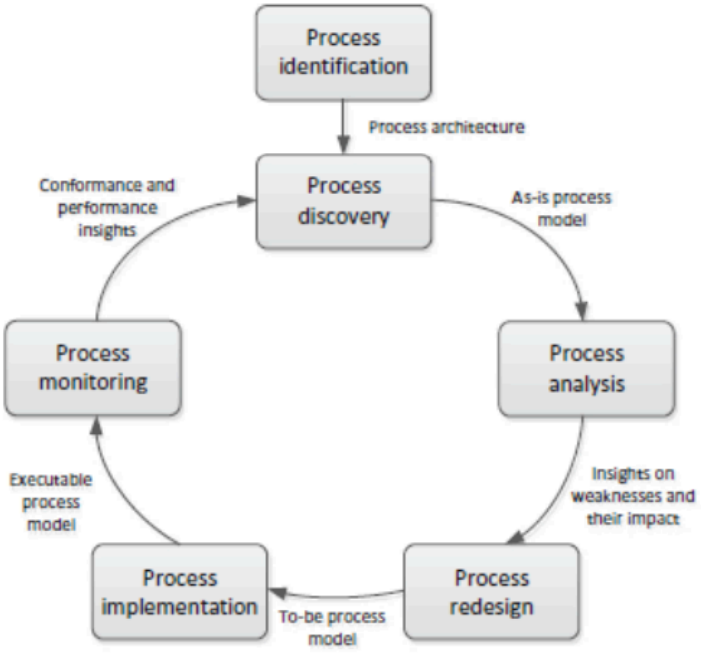
Stakeholders in BPM Lifecycle

- **Management Team:**
  - **CEO:** Oversees overall business performance (Example: CEO of Amazon deciding business strategy).
  - **COO:** Manages how operations run (Example: COO of a manufacturing company optimizing production lines).
  - **CPO:** Focuses on process performance (Example: CPO ensuring order fulfillment process is smooth).
  - **CIO:** Ensures IT systems work efficiently (Example: CIO managing ERP systems at a bank).
  - **CFO:** Manages company finances (Example: CFO monitoring budgets in a retail chain).

    - **HRD:** Manages employees involved in processes (Example: HRD organizing training for factory workers).
- **Process Owners:**
    - Responsible for the performance of a specific process.
    - Plan, monitor, and guide the process (Example: Owner of the procurement process ensuring timely purchase of materials).
- **Process Participants:**
    - Employees who carry out daily tasks in a process (Example: Cashiers processing sales at a supermarket).
- **Process Analysts/Supervisors:**
    - Identify, analyze, and improve processes.
    - Coordinate implementation and monitor performance (Example: Analyst redesigning a loan approval process in a bank).
- **System Engineers:**
    - Help redesign processes in IT systems.
    - Work with analysts to capture system requirements (Example: IT engineer implementing ERP workflow for HR approvals)
- **BPM Team (BPM Center of Excellence)**
    - Designs, implements, and improves business processes to meet company goals.
    - Ensures processes are efficient, effective, and flexible for changes.
    - Keeps process documents updated and organized.
    - Plans and manages BPM projects.
    - Maintains overall process structure and decides which processes need improvement first.
    - Supports other stakeholders like process owners, analysts, and participants.

Example: At a large bank, the BPM team ensures that loan processing, account opening, and customer service workflows are smooth, documented, and updated regularly to meet regulatory requirements and customer expectations.

BPM life cycle



BPM Life Cycle

1. **Process Identification**
   Recognize and prioritize processes that need improvement. Map existing processes, identify key processes, define boundaries, inputs, outputs, and stakeholders.

   Example: A bank identifies loan approval and account opening as critical processes that need efficiency improvements.

2. **Process Discovery (As-Is Analysis)**
   Understand the current state of the process. Document workflows with diagrams, gather data from employees and stakeholders, and spot inefficiencies or bottlenecks.

   Example: The bank maps the loan approval process and finds delays due to manual document verification.

3. **Process Analysis**
   Examine the as-is process to find improvement opportunities. Use tools like root cause analysis and value stream mapping. Evaluate performance against goals and identify automation or redesign needs.

   Example: The bank discovers that automating document checks can reduce loan

approval time by 50%.

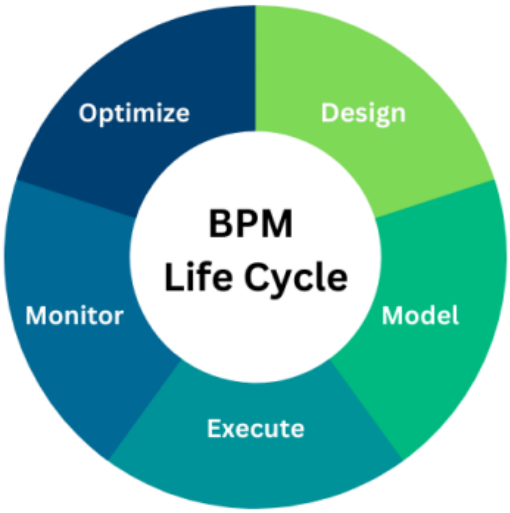
4. **Process Redesign (To-Be Design / Redesign)**
   Create an improved process that fixes inefficiencies and meets business goals. Incorporate technology, automation, and best practices. Define roles, responsibilities, and workflows.

   Example: The bank implements a digital document verification system and redesigns staff roles to speed up loan approvals.

5. **Process Implementation**
   Deploy the new process. Introduce tools, train employees, and run pilot tests before full rollout.
   Example: The bank runs a pilot with one branch before launching the new loan process across all branches.

6. **Process Monitoring and Controlling**
   Track performance using KPIs and dashboards. Collect feedback and compare results with goals.
   Example: The bank monitors loan approval times and customer satisfaction to ensure the process works efficiently.

7. **Process Optimization (Continuous Improvement)**
   Continuously improve the process using feedback and performance data. Apply methods like Lean, Six Sigma, or PDCA. Adapt to changing business needs or technology.
   Example: The bank further optimizes the loan process by adding AI-based credit scoring to speed approvals and reduce errors.



(Lecture 02)-: Process Identification

Process Identification

- Management activities to clearly define all business processes and set rules for choosing which to improve.
  Example: A bank maps processes like account opening, loan approval, and fund transfer, then selects loan approval for improvement.

Focusing on Key Processes

- Some processes are more important for the organization's success and need priority.
- Focus on areas with high value or big problems.
- Need to answer two questionsAsk:

  1) What processes exist?

  2) Which should we focus on?

- Two phases:
  - **Designation:** Understand all processes and how they connect.
  - **Evaluation:** Prioritize processes for management activities like redesign or automation.

Example: An e-commerce company maps all processes like order processing, delivery, and returns. It then focuses on delivery because delays affect customer satisfaction the most.

The Designation Phase

- Identify all processes in the organization.
- Map how processes are connected
- Document details of each process.
- Define where each process starts and ends.
- Analyze how well each process currently performs.

  Example: A bank lists all processes like account opening, loan approval, and fund transfer, maps how they connect, and checks current efficiency before improving them.

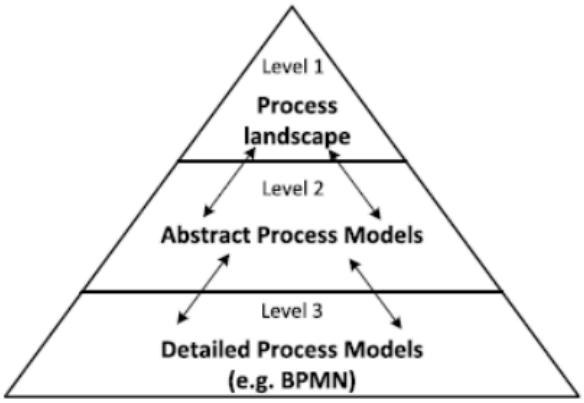
The Evaluation Phase

• Check how well each process is performing.
• Decide which processes need improvement first.
• Select the right process management activities (like redesign, automation).

• Assess if improvements are possible with available resources.

Example: A retail company evaluates checkout, inventory, and delivery processes, prioritizes improving slow delivery first, and checks if they have enough staff and technology to make changes.

Fig. 2.1 The different levels of detail in a process architecture



PROCESS CHECKLIST

• Is it a process at all?
• Can the process be controlled?
• Is the process important enough to manage?
• Is the scope of the process not too big?
• Is the scope of the process not too small?

Why Process Architecture.

- Shows all the processes in an organization clearly.
- Helps manage the complexity of the entire organization.
- **First**, categorize different types of processes; **second**, explain how they are connected.

  Example: A bank creates a process architecture showing categories like account opening, loan processing, and customer service, and maps how account opening links to customer service and compliance checks.

Level 01: Process Categories for Landscaping

Uses Porter's Value Chain model

Core Processes

These are the **main activities** that directly create value for customers.

Examples inside a company:

- Designing products
- Manufacturing
- Marketing & sales
- Delivery
- After-sales service
- Direct procurement (buying materials needed for the main product)

  example:
  A restaurant's core processes are cooking the food, serving customers, and buying ingredients.

Support Processes

These help the core processes work smoothly, but they **do not directly create customer value**.

Examples:

- HR (hiring staff)
- IT support
- Accounting
- Financial management
- Legal services
  Indirect procurement (buying office items like chairs, paper, etc.)

  example:
  In a school, support processes include hiring teachers, maintaining computers, and doing accounts.

Management Processes

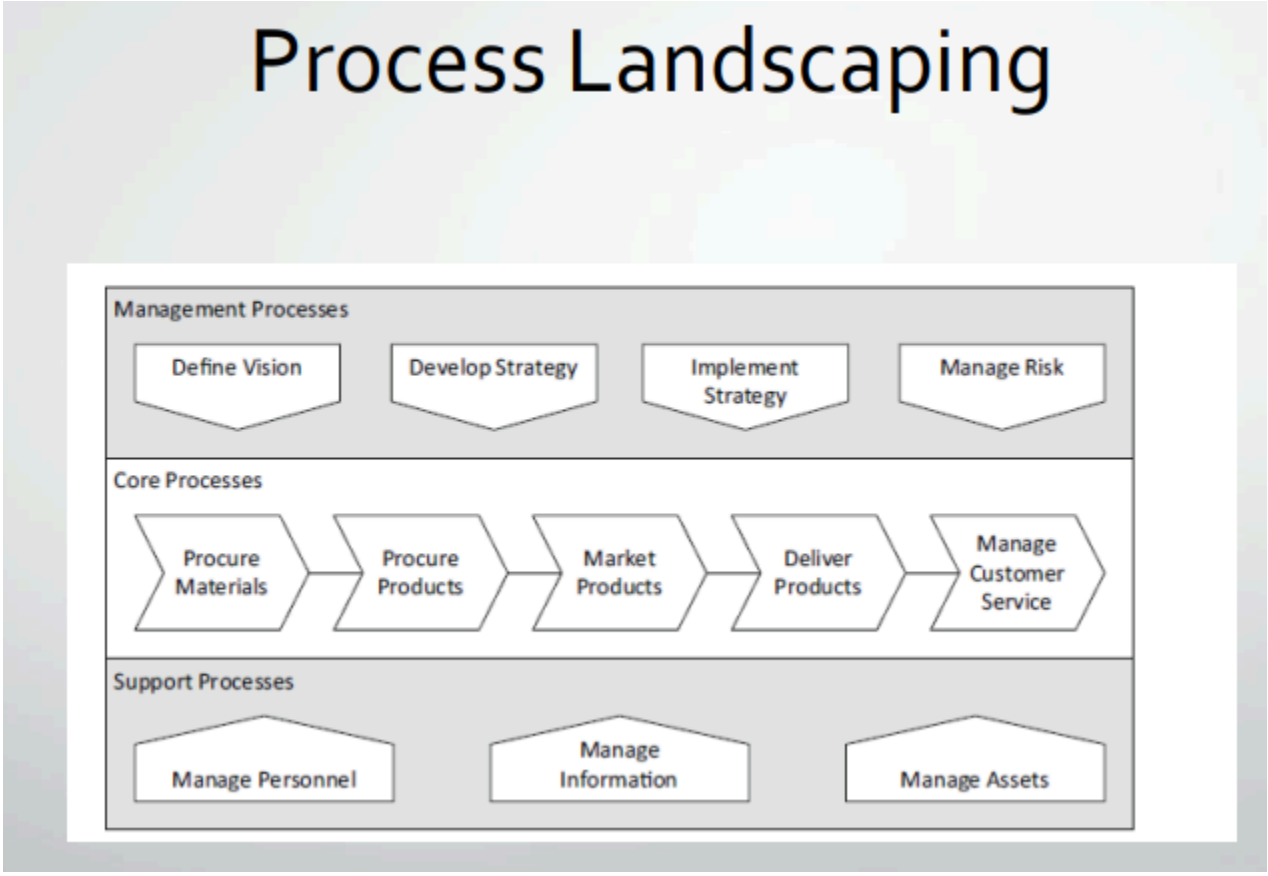
These are processes that guide and control the whole organization. Done by managers, leaders, focus on planning and decision making, monitor performance

Examples:

- Strategic planning
- Budgeting
- Compliance & risk management
- Managing investors, suppliers, and partners

example:
In a hospital, management processes include planning budgets, setting goals, and managing supplier contracts.

Example: A coffee shop's core processes include making and serving coffee to customers. Support processes include hiring staff and maintaining the coffee machines. Management processes include planning budgets, setting goals, and managing supplier contracts.



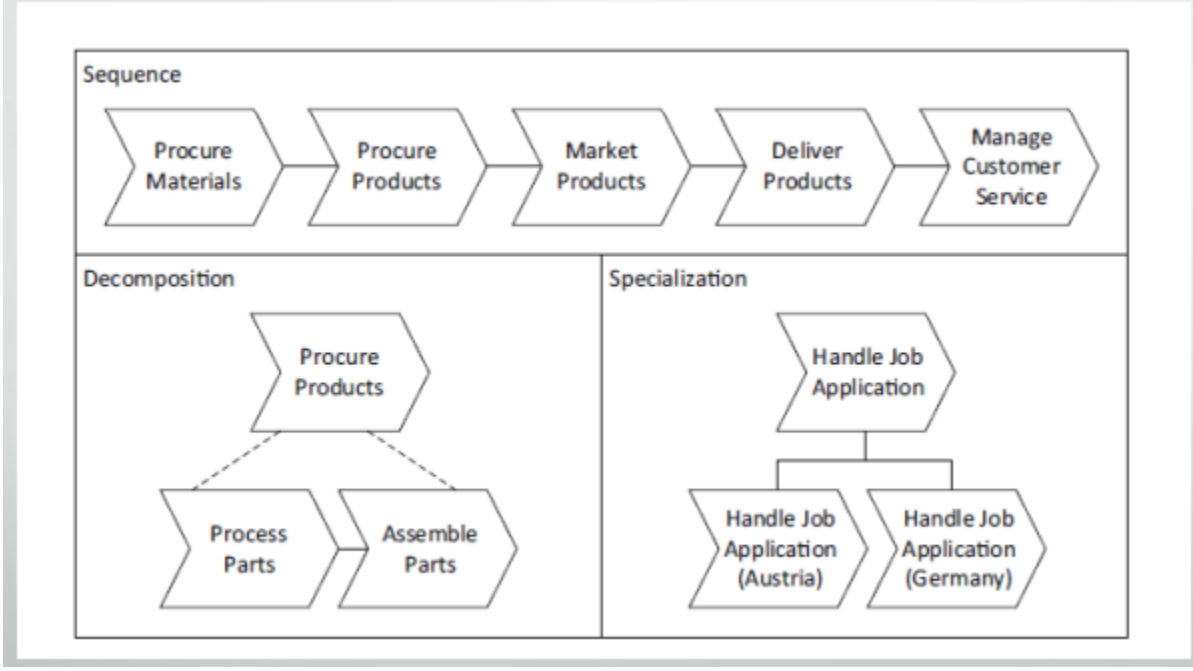
Relationships Between Processes

• **Sequence:** The order in which processes are carried out.

• **Decomposition:** Breaking a process into smaller, manageable sub-processes.

• **Specialization:** Assigning specific tasks or processes to specialized teams or roles.

Example: In a landscaping company, first the design process (sequence), then it is broken into planting and irrigation tasks (decomposition), handled by specialized teams like gardeners and irrigation specialists (specialization).



Reuse of Reference Models

- Process analysts often find it hard to identify all processes and layers of a process architecture.
- Using reference models can help as a guide.
- These models are developed by industry groups, government, non-profits, and universities.

Examples of some frameworks:

- Information Technology Infrastructure Library (ITIL)
- Supply Chain Operations Reference Model (SCOR)
- Process Classification Framework
- Performance Framework

Example: A manufacturing company can use SCOR to map and improve its supply chain processes instead of starting from scratch.

Process landscape model of Vienna's public transport operator

(example)

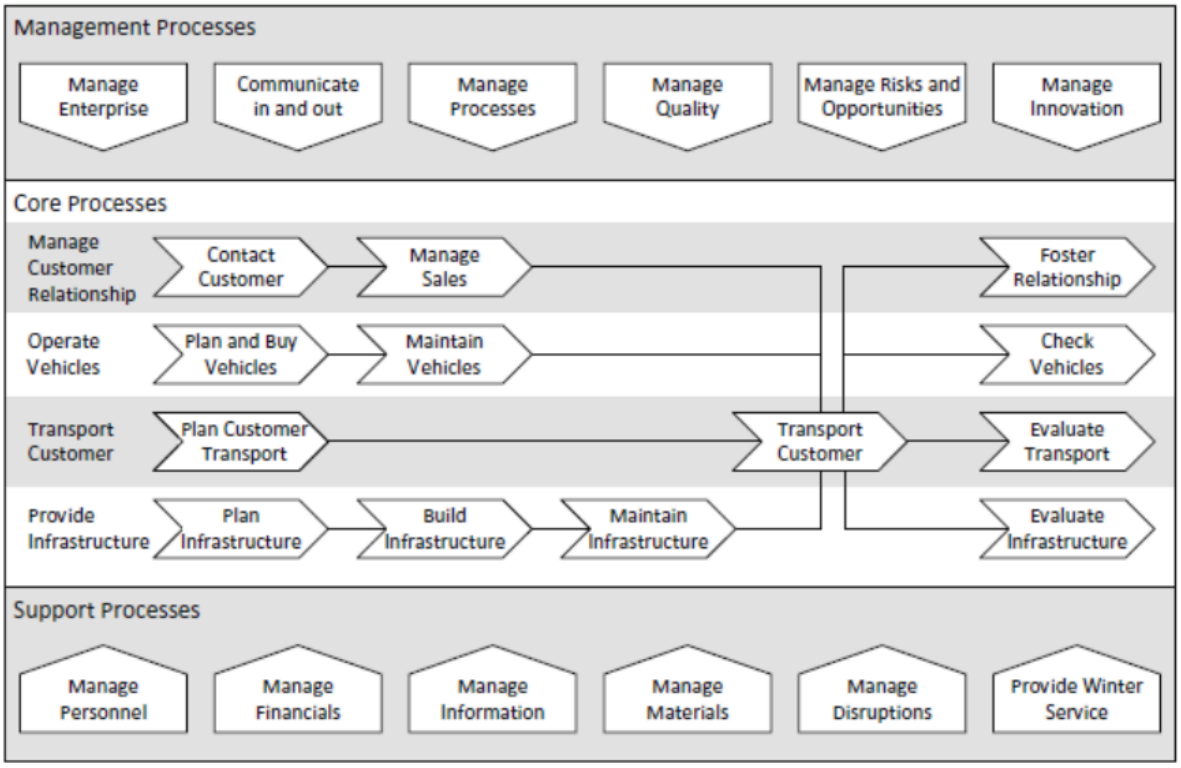


Table 2.1 Level 1 and Level 2 of the APQC Process Classification Framework

1.0 Develop Vision and Strategy
1.1 Define the business concept and long-term vision
1.2 Develop business strategy
1.3 Execute and measure strategic initiatives

2.0 Develop and Manage Products and Services
2.1 Govern and manage product and service development program
2.2 Generate and define new product and service ideas
2.3 Develop products and services

3.0 Market and Sell Products and Services
3.1 Understand markets, customers, and capabilities
3.2 Develop marketing strategy
3.3 Develop and manage marketing plans
3.4 Develop sales strategy
3.5 Develop and manage sales plans

4.0 Deliver Physical Products
4.1 Plan for and align supply chain resources
4.2 Procure materials and services
4.3 Produce, manufacture, and deliver product
4.4 Manage logistics and warehousing

5.0 Deliver Services
5.1 Establish service delivery governance and strategies
5.2 Manage service delivery resources
5.3 Deliver service to customer

6.0 Manage Customer Service
6.1 Develop customer care and customer service strategy
6.2 Plan and manage customer service contacts
6.3 Service products after sales
6.4 Manage product recalls and regulatory audits
6.5 Evaluate customer service operations and customer satisfaction

7.0 Develop and Manage Human Capital
7.1 Develop and manage human resources planning, policies, and strategies
7.2 Recruit, source, and select employees
7.3 Develop and counsel employees
7.4 Manage employee relations
7.5 Reward and retain employees
7.6 Redeploy and retire employees
7.7 Manage employee information and analytics
7.8 Manage employee communication
7.9 Deliver employee communications

8.0 Manage Information Technology (IT)
8.1 Manage the business of information technology
8.2 Develop and manage IT customer relationships
8.3 Develop and implement security, privacy, and data protection controls
8.4 Manage enterprise information
8.5 Develop and maintain information technology solutions
8.6 Deploy information technology solutions
8.7 Deliver and support information technology services

9.0 Manage Financial Resources
9.1 Perform planning and management accounting
9.2 Perform revenue accounting
9.3 Perform general accounting and reporting
9.4 Manage fixed-asset project accounting
9.5 Process payroll
9.6 Process accounts payable and expense reimbursements
9.7 Manage treasury operations
9.8 Manage internal controls
9.9 Manage taxes
9.10 Manage international funds/consolidation
9.11 Perform global trade services

10.0 Acquire, Construct, and Manage Assets
10.1 Plan and acquire assets
10.2 Design and construct productive assets
10.3 Maintain productive assets
10.4 Dispose of assets

11.0 Manage Enterprise Risk, Compliance, Remediation and Resiliency
11.1 Manage enterprise risk
11.2 Manage compliance
11.3 Manage remediation efforts
11.4 Manage business resiliency

12.0 Manage External Relationships
12.1 Build investor relationships
12.2 Manage government and industry relationships
12.3 Manage relations with board of directors
12.4 Manage legal and ethical issues
12.5 Manage public relations program

13.0 Develop and Manage Business Capabilities
13.1 Manage business processes
13.2 Manage portfolio, program, and project
13.3 Manage quality
13.4 Manage change
13.5 Develop and manage enterprise-wide knowledge management (KM) capability
13.6 Measure and benchmark
13.7 Manage environmental health and safety (EHS)

Steps for Model Landscaping

- Clarify terminology used in processes.
- Identify end-to-end processes across the organization.
- For each end-to-end process, list the sequence of its subprocesses.
- For each business process, identify key management and support processes.
- Break down (decompose) and specialize processes as needed.
- Compile a process profile for documentation.
- Check that all processes are complete and consistent.

Example: In a restaurant, start with “Order to Delivery” as an end-to-end process, list cooking, packing, and delivery as sequential processes, then include HR, inventory, and accounting as support processes.

Name of Process: Procure-to-Pay	
Vision: The objective of the procurement process is to secure that the entire range of external products and services becomes available on time and is at the required level of quality.	
Process Owner: Chief Financial Officer (CFO)	
Customer of process: • Requesting unit	**Expectation of customer:** • Timely, economic and complete provision
Outcome: Delivered products or provided services for the requested unit	
Trigger: Need is identified	
First activity: Submit Request ..... **Last activity:** Create Purchase Order	
Interfaces inbound: Plan-to-Procure **Interfaces outbound:** Construct-to-Complete	
Required resources: • Human resources: Site Engineer, Clerk, Works Engineer • Information, documents, know-how: procurement guidelines, supplier rating, framework contract • Work environment, materials, infrastructure: Procurement information system	
Process Performance Measures: • Cycle Time • Operational Costs • Error Rate	

Selecting and Prioritizing

Selection Criteria

- Strategic Importance – How critical the process is for business goals.
- Health – Current performance and issues in the process.
- Feasibility – Ease of improving or changing the process.

Process Performance Measures

- Use key performance indicators (KPIs) to evaluate processes.
- Four dimensions:
    - Time – speed of the process
    - Cost – expenses involved
    - Quality – accuracy and standard of output
    - Flexibility – ability to adapt to changes

Example: For an e-commerce company, the order fulfillment process might be prioritized because it is strategic, needs improvement in speed (time) and accuracy (quality), and is feasible to optimize with automation.

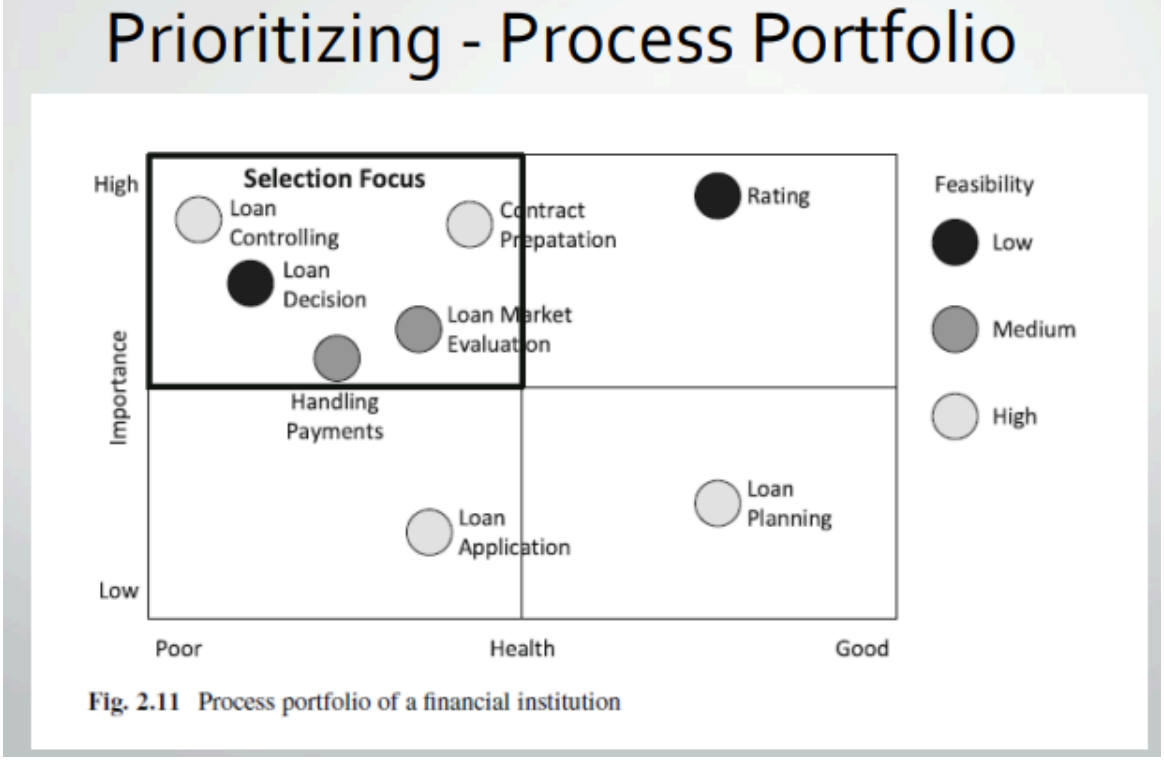
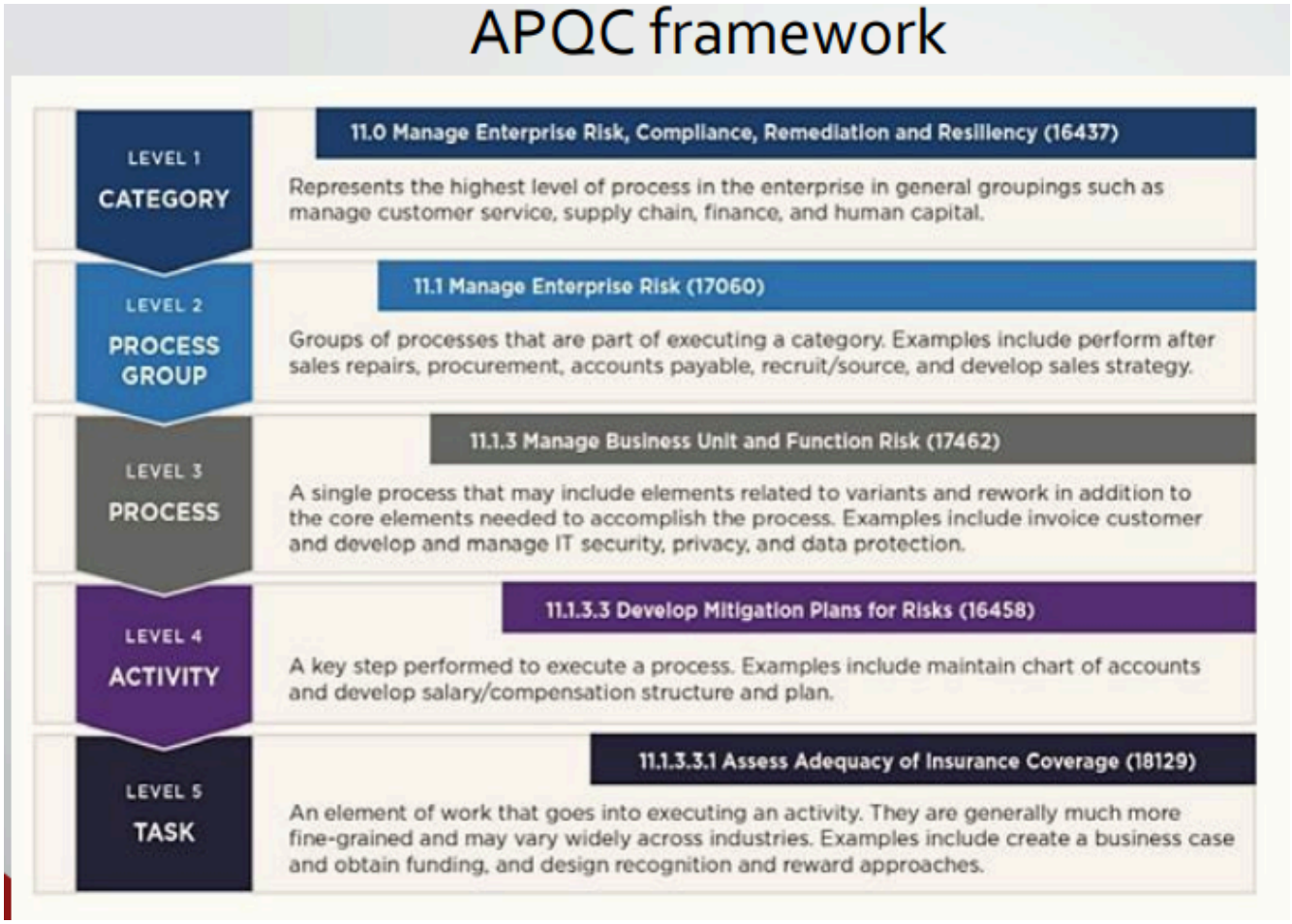
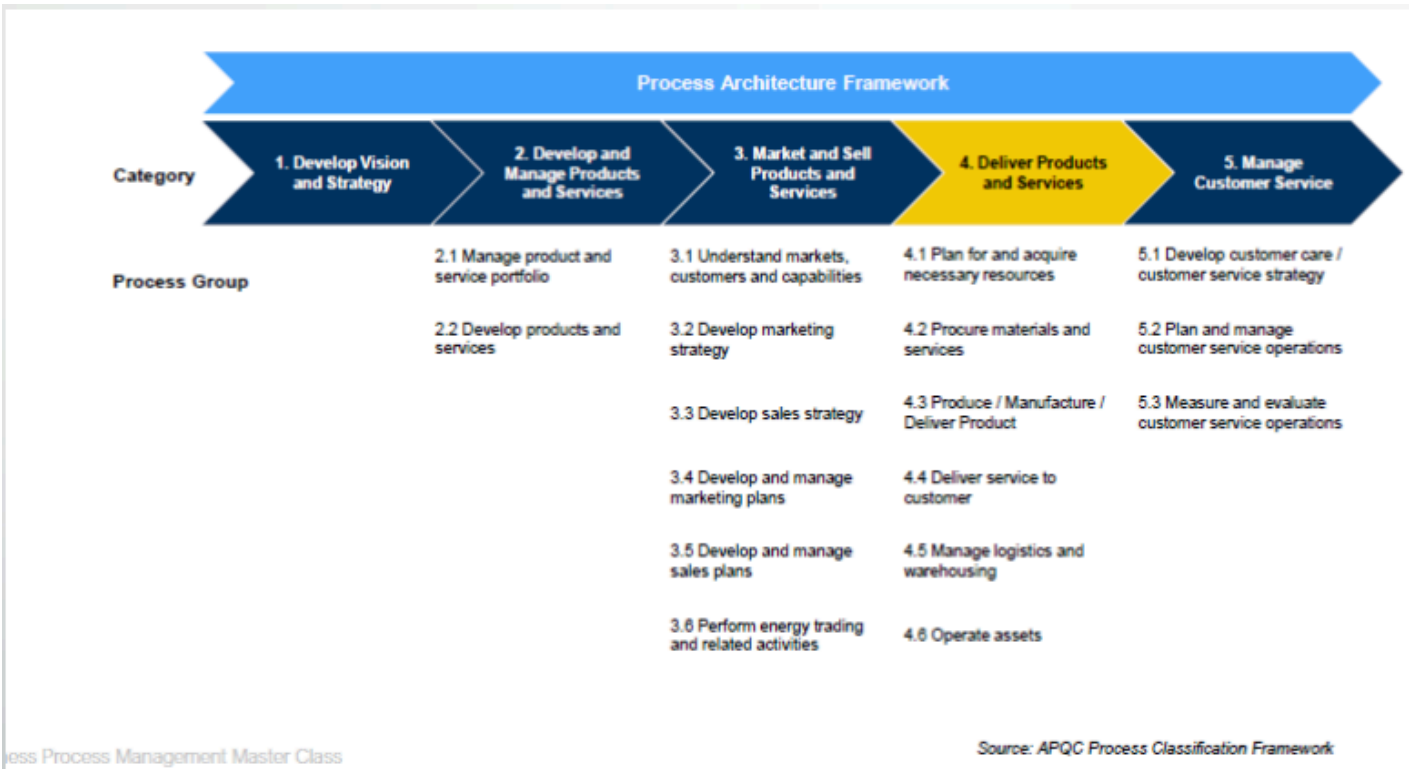


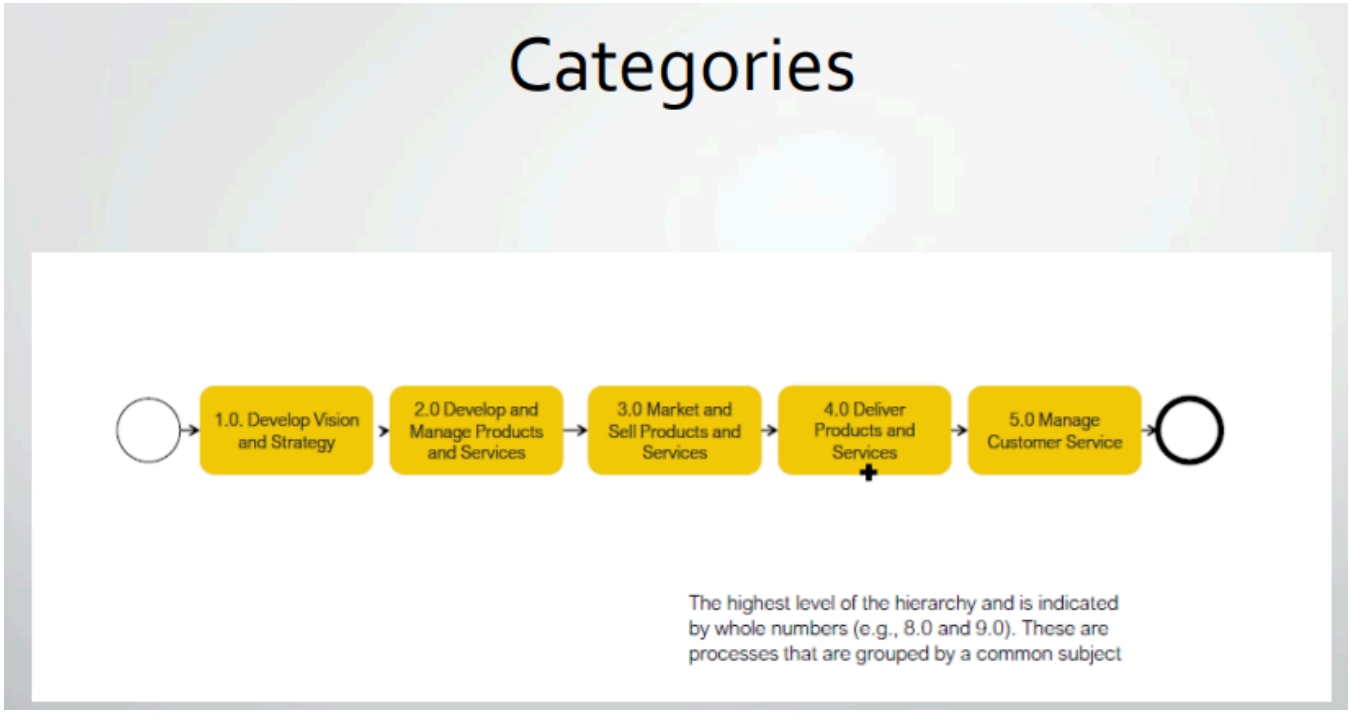
Fig. 2.11 Process portfolio of a financial institution

APQC framework

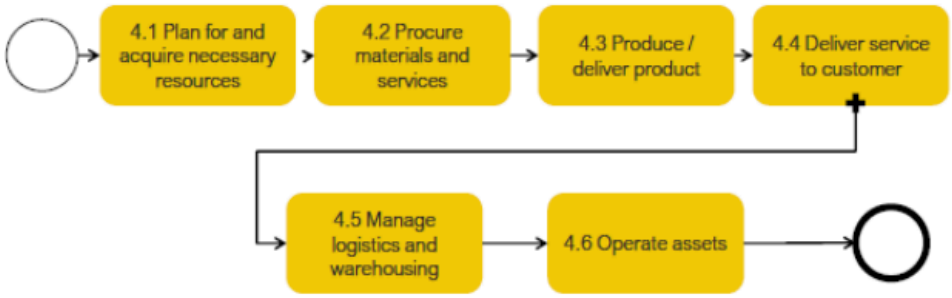


Example:

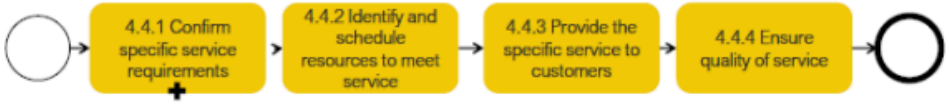

Process Architecture Framework
Category
1. Develop Vision and Strategy
2. Develop and Manage Products and Services
3. Market and Sell Products and Services
4. Deliver Products and Services
5. Manage Customer Service
Process Group
2.1 Manage product and service portfolio
2.2 Develop products and services
3.1 Understand markets, customers and capabilities
3.2 Develop marketing strategy
3.3 Develop sales strategy
3.4 Develop and manage marketing plans
3.5 Develop and manage sales plans
3.6 Perform energy trading and related activities
4.1 Plan for and acquire necessary resources
4.2 Procure materials and services
4.3 Produce / Manufacture / Deliver Product
4.4 Deliver service to customer
4.5 Manage logistics and warehousing
4.6 Operate assets
5.1 Develop customer care / customer service strategy
5.2 Plan and manage customer service operations
5.3 Measure and evaluate customer service operations
Source: APQC Process Classification Framework

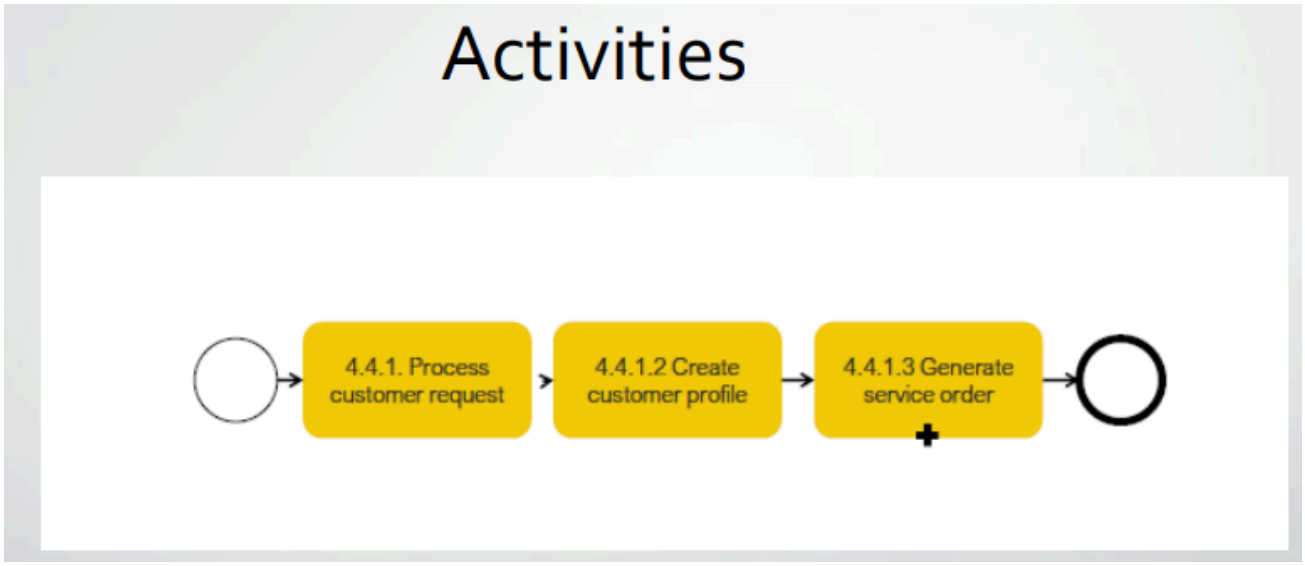

Categories
1.0. Develop Vision and Strategy
2.0 Develop and Manage Products and Services
3.0 Market and Sell Products and Services
4.0 Deliver Products and Services
5.0 Manage Customer Service
The highest level of the hierarchy and is indicated by whole numbers (e.g., 8.0 and 9.0). These are processes that are grouped by a common subject

Process Grouping



Process







Exercise 2.1 • Develop a University Process Architecture for UWU undergraduate (From admission to graduation (convocation)).

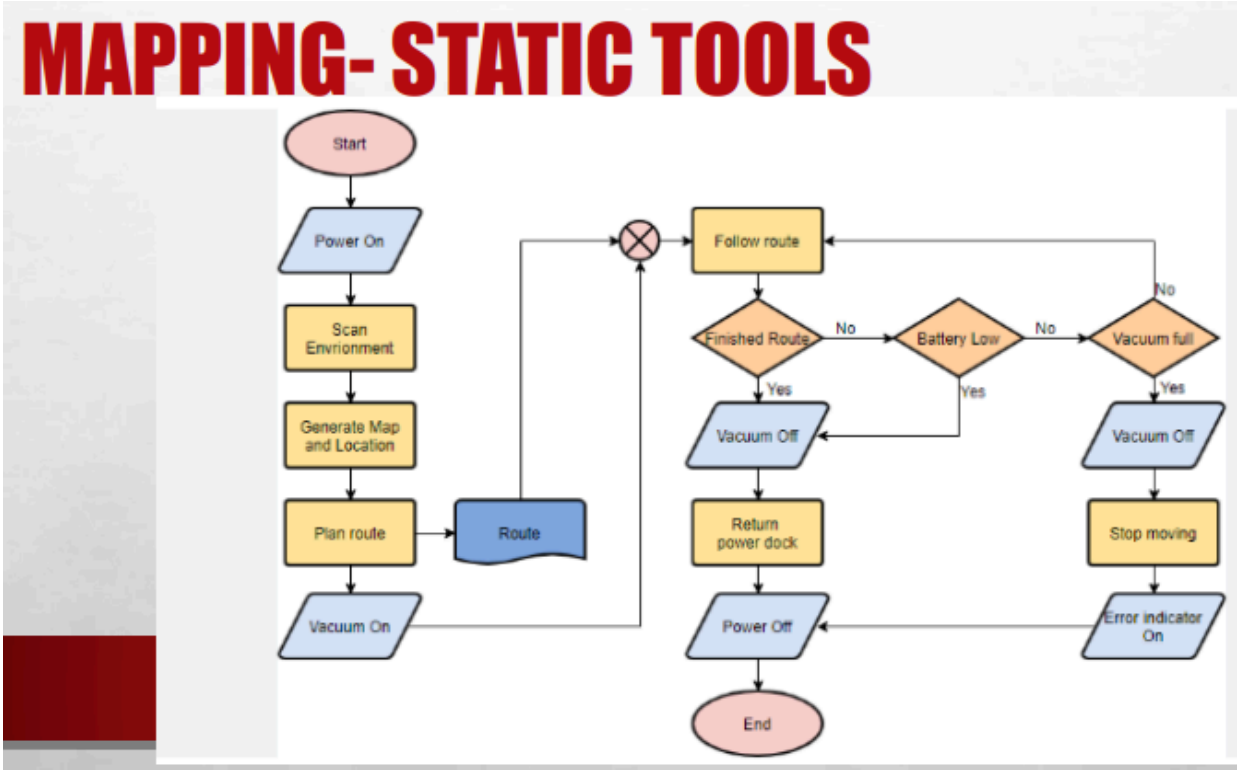
Figure 1: Dataset Preparation (Red)

Lecture 3 : BUSINESS PROCESS MANAGEMENT PROCESS MODELING

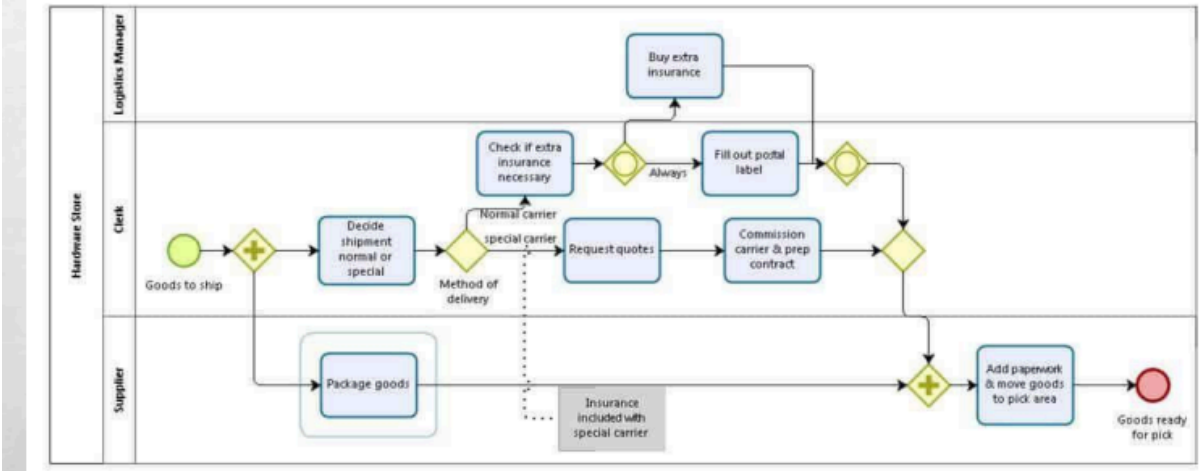
WHY PROCESS MODELING?

• Helps understand the process and share that understanding with people involved daily.
• Allows identification and prevention of issues in the process.
• Provides a clear understanding of business processes, which is needed before analyzing, redesigning, or automating them.

Example: A bank models its loan approval process so all staff know the steps, spot bottlenecks, and prepare for automation using software.



MODELING – DYNAMIC TOOLS



ARIS SOFTWARE TOOL

- ARIS is a software used for modeling, analyzing, and improving business processes.
- Helps visualize processes clearly using diagrams and flowcharts.
- Supports process documentation, analysis, and simulation for better decision-making.

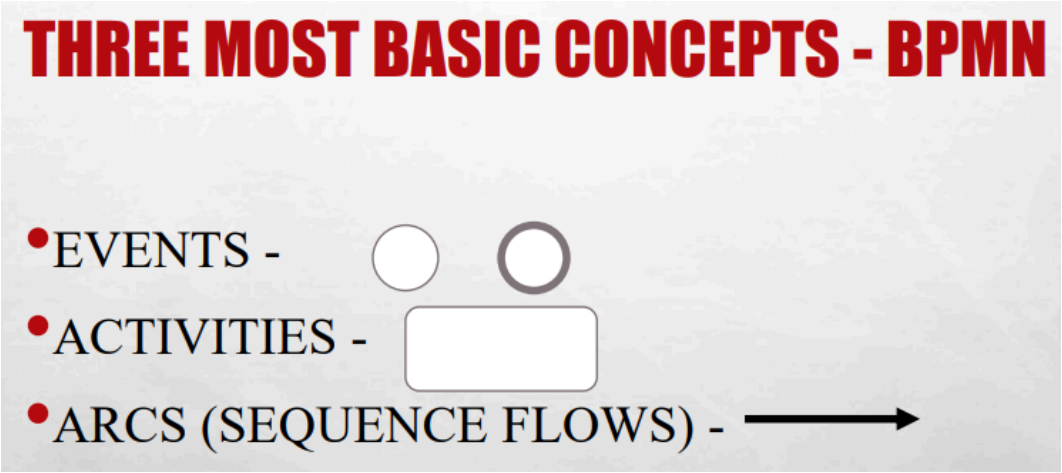


DIFFERENT MODELING METHODS

- Business Process Model And Notation (BPMN)
- Unified Modeling Language (UML)
- Flowchart

COMPONENTS OF A MODELING LANGUAGE

- includes
- activities,
- events,
- gateways,
- sequence flows



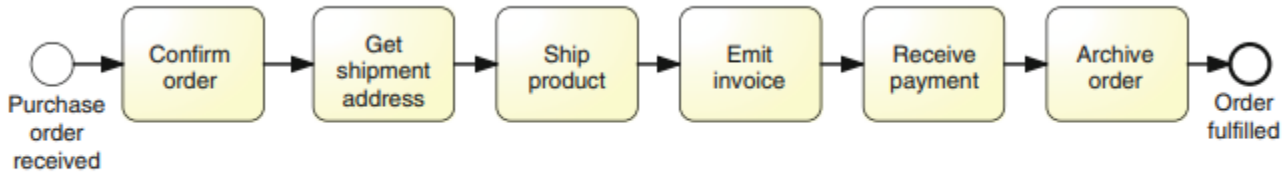
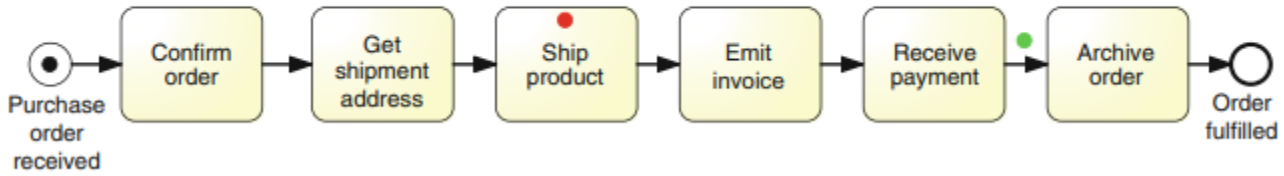


Fig. 3.1 The model of a simple order-to-cash process



NAMING CONVENTIONS

- **Activities:** Start with a verb and capitalize the first letter.
  Example: Approve Order
- **Events:** Start with a noun and capitalize the first letter.
  Example: Payment Received
- **Name a Process Model:** Use a noun, sometimes with an adjective.
  Example: Loan Origination, Order-to-Cash

BRANCHING AND MERGING

Processes can split or run at the same time.

Gateway: Shows where a process splits or joins.

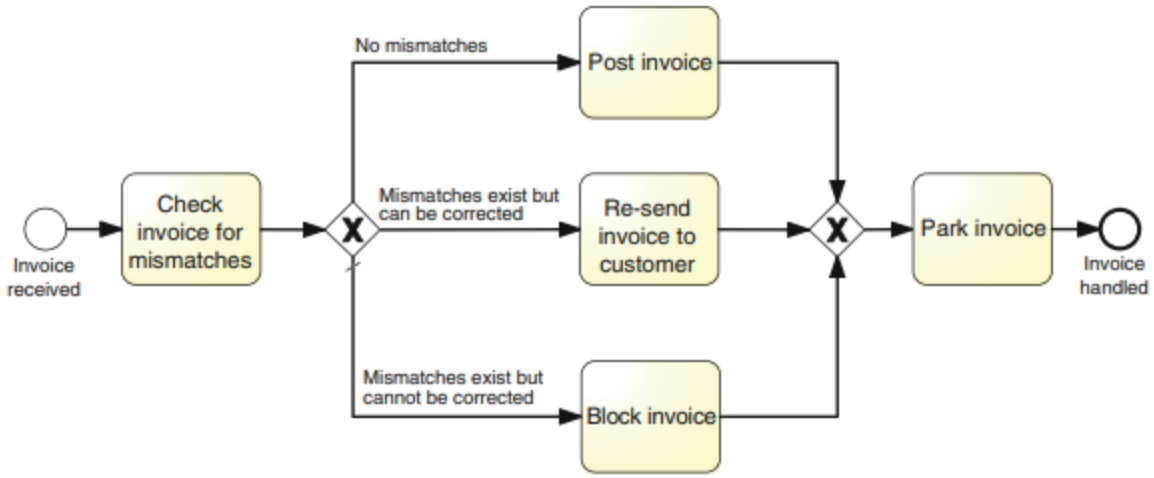
- **Split Gateway:** Process diverges.
- **Join Gateway:** Process converges.

A split gateway represents a point where the process flow diverges while a join gateway represents a point where the process flow converges.

BRANCHING AND MERGING - EXCLUSIVE DECISIONS (XOR) කිහිපයකින් එකක්

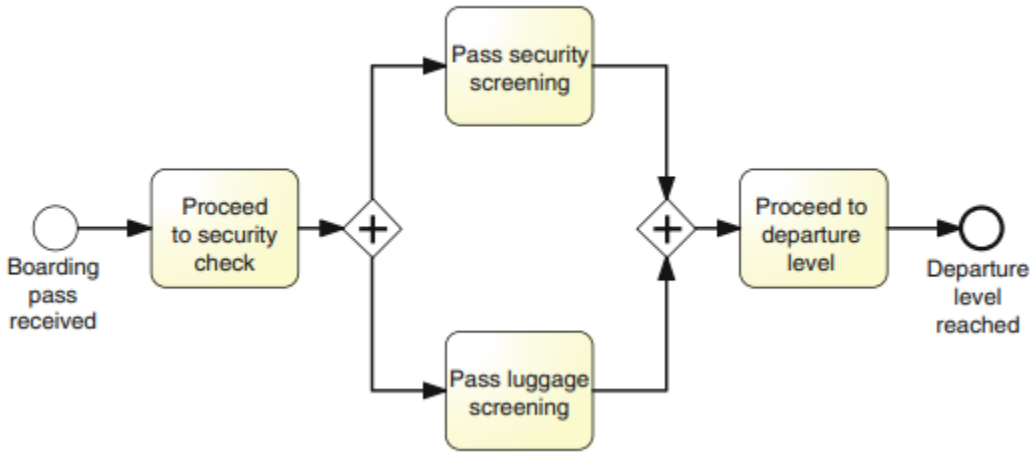
As soon as an invoice is received from a customer, it needs to be checked for mismatches. The check may result in any of the following three options:

(i) there are no mismatches, in which case the invoice is posted;
(ii) there are mismatches but these can be corrected, in which case the invoice is resent to the customer;
(iii) there are mismatches but these cannot be corrected, in which case the invoice is blocked.
Once one of these three activities is performed the invoice is parked and the process completes.

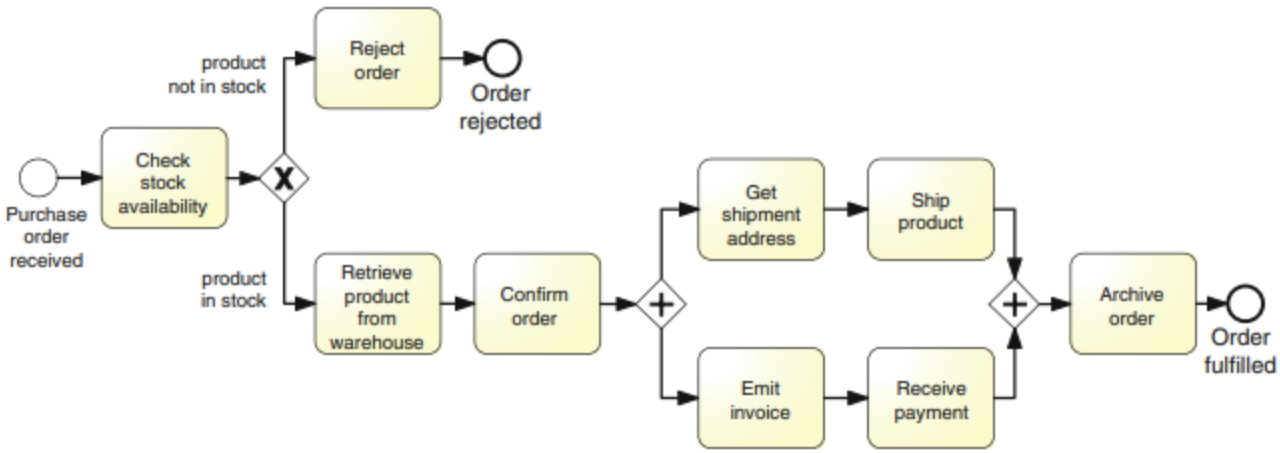


BRANCHING AND MERGING - PARALLEL EXECUTION (AND) කිහිපයකින් ඔක්කොම

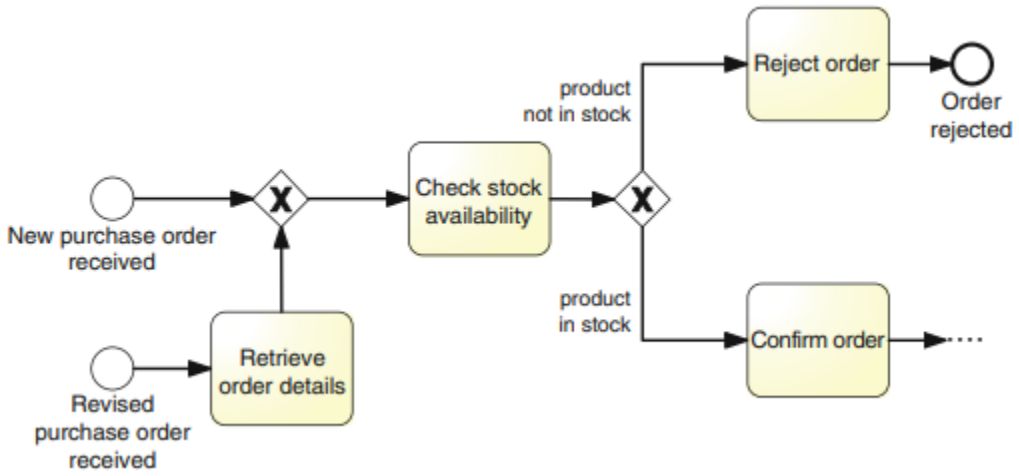
Once the boarding pass has been received, passengers proceed to the security check. Here they need to pass the personal security screening and the luggage screening. Afterwards, they can proceed to the departure level.



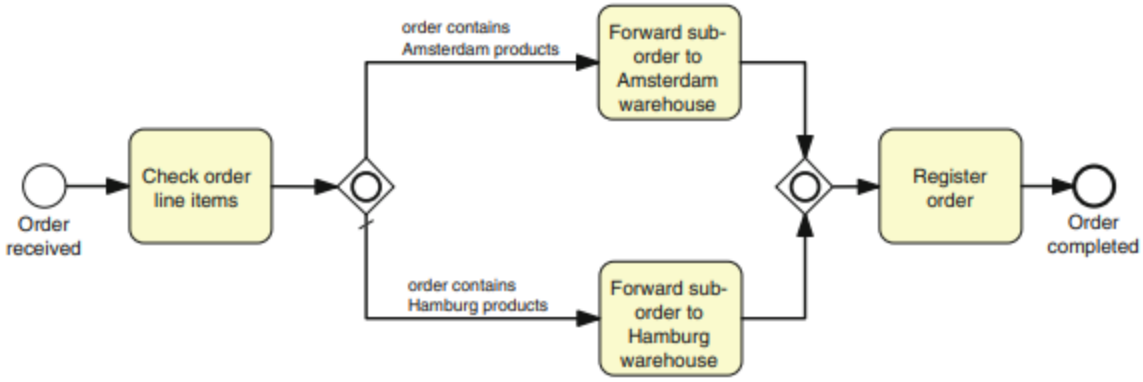
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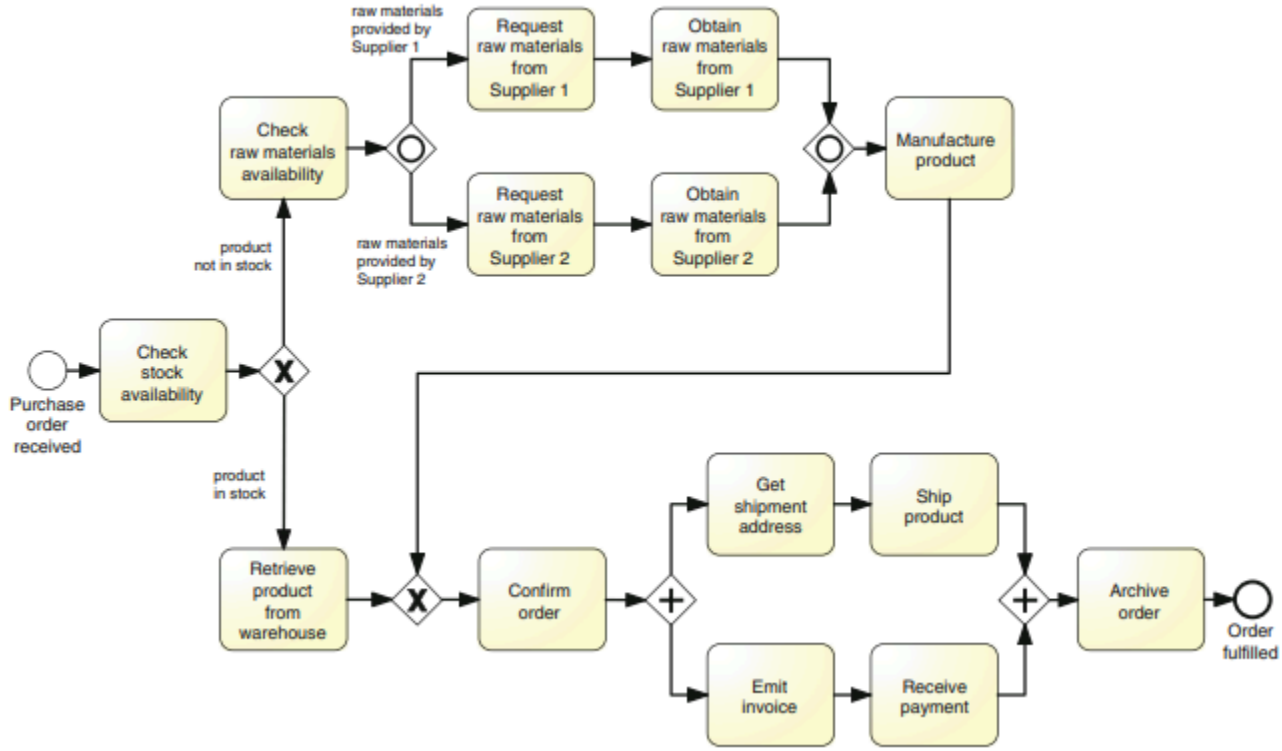
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BRANCHING AND MERGING - INCLUSIVE DECISIONS (OR) දෙකම හෝ දෙකෙන් එකක්

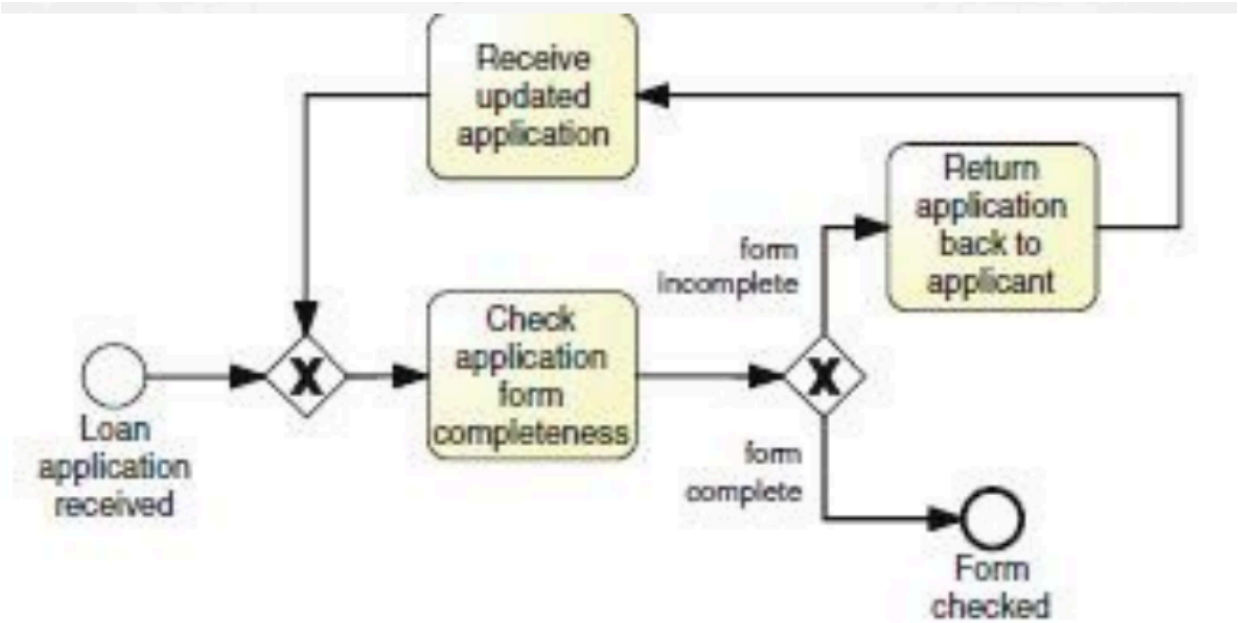


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BRANCHING AND MERGING - REWORK AND REPETITION

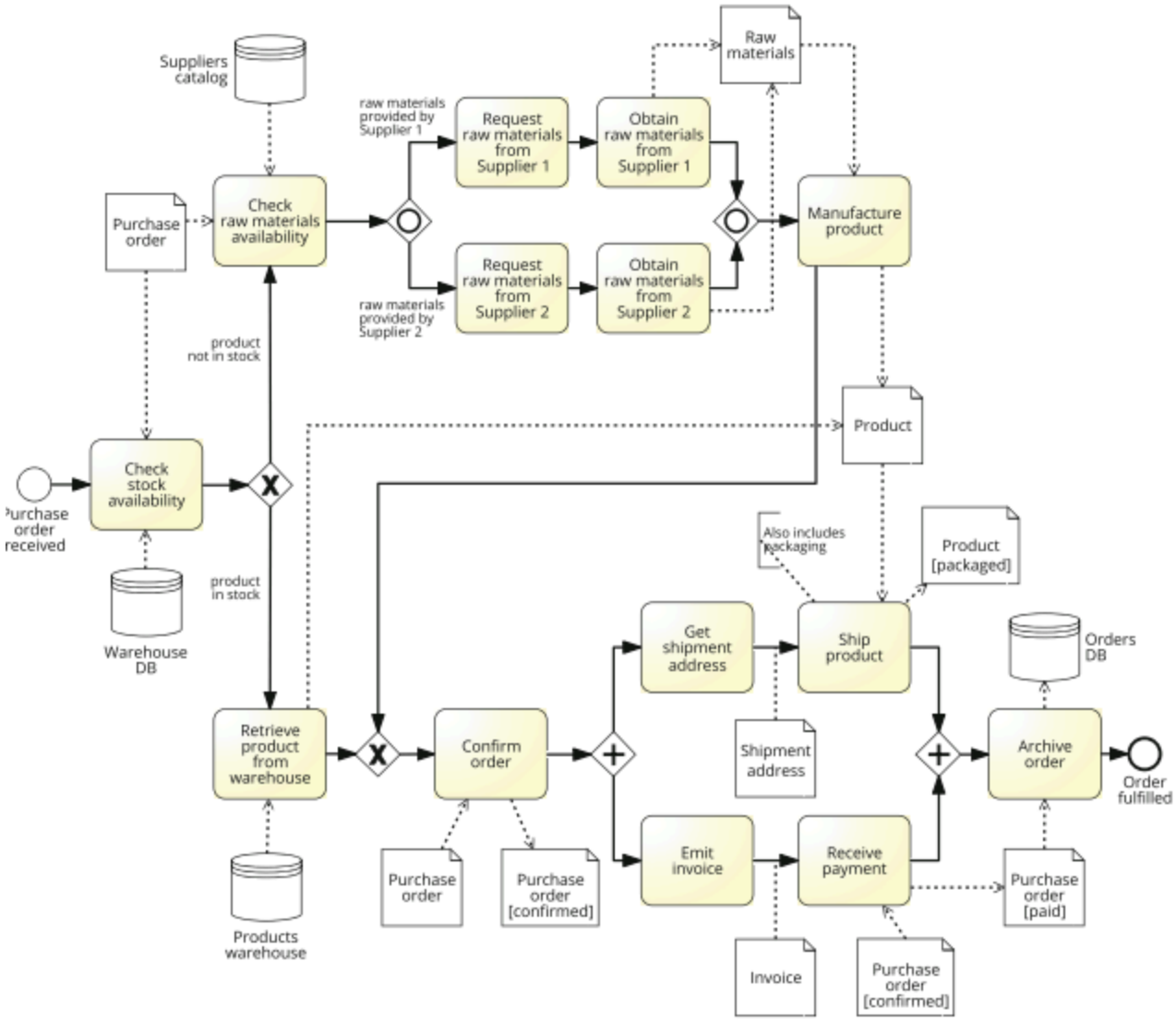
Once a loan application is received by the loan provider, and before proceeding with its assessment, the application itself needs to be checked for completeness. If the application is incomplete, it is returned to the applicant, so that they can fill out the missing information and send it back to the loan provider. This process is repeated until the application is found complete.



BUSINESS OBJECTS

- The **object perspective** (or **data perspective**) shows which business objects are used or created in a process.
- **Business objects** (or artifacts) can be documents, files, materials, etc.
- The **direction of data** shows if the object is an input or output for an activity.

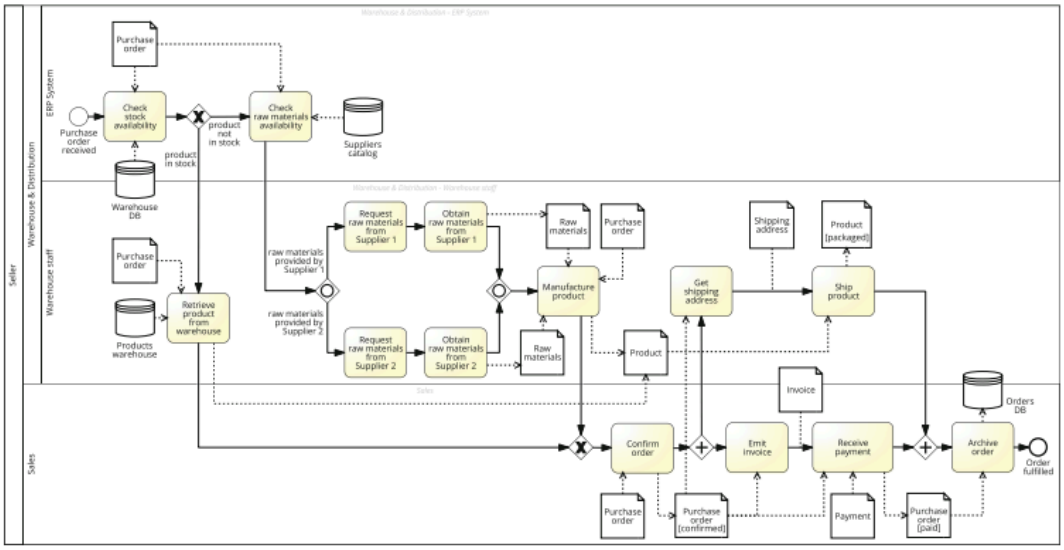
Example: In a purchase order process, a "Purchase Request" document is an input to the "Approve Order" activity, and an "Approved Order" document is the output.



RESOURCE

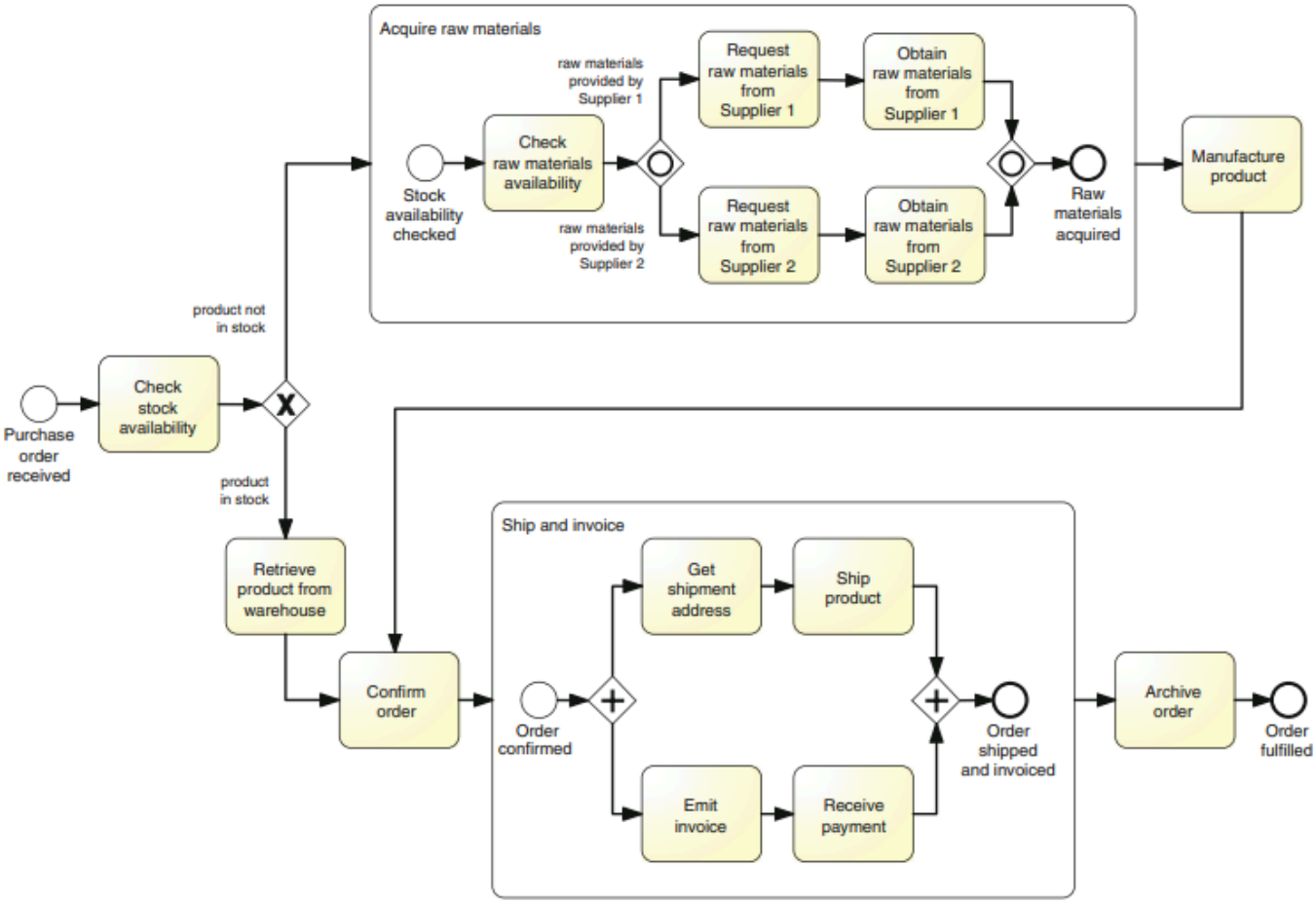
- A process participant, i.E. An individual person like the employee john smith,
- A software system, for example a server or a software application,
- A piece of equipment, such as a printer or a manufacturing plant.

Fig. 3.15 The order-to-cash example with resource information

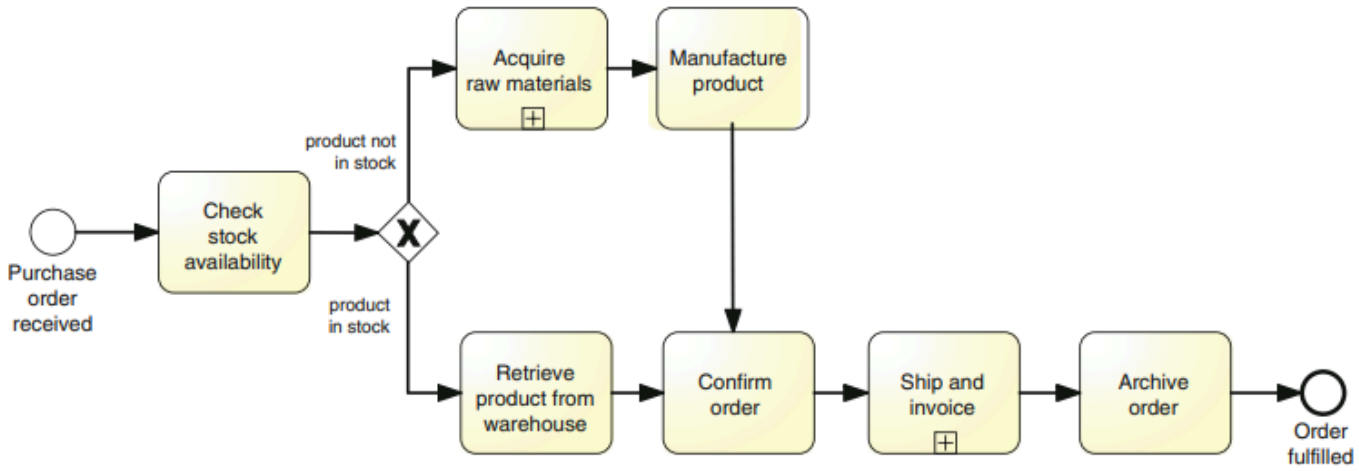


Process Decomposition – Simple Explanation

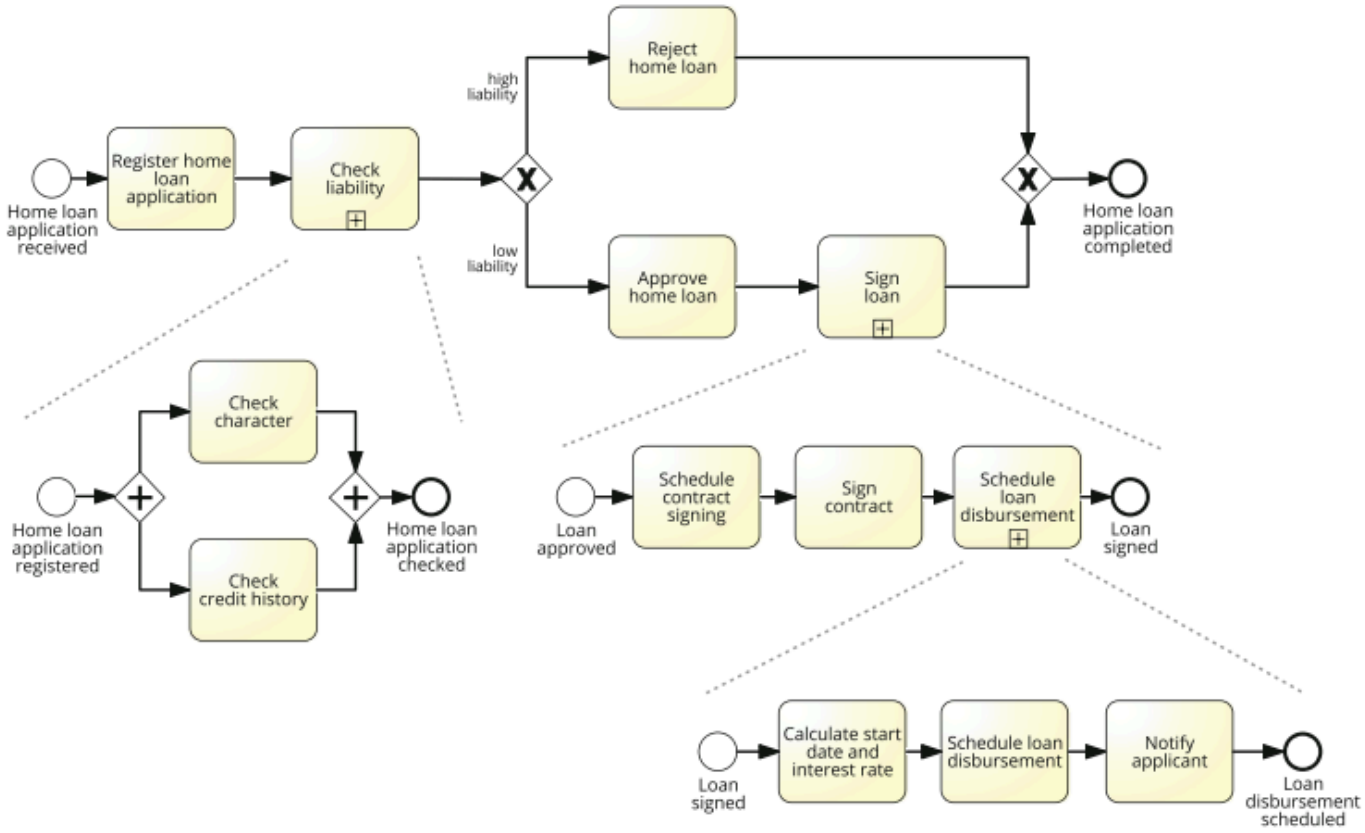
- We break a big process into smaller parts to make it easier to understand.
- Some parts can be grouped together into a **sub-process**.
- A **task** (atomic activity) is the smallest activity that cannot be divided further.
- To create a sub-process, we first find activities that are related and group them together.


Acquire raw materials
raw materials provided by Supplier 1
Request raw materials from Supplier 1
Obtain raw materials from Supplier 1
Check raw materials availability
Stock availability checked
raw materials provided by Supplier 2
Request raw materials from Supplier 2
Obtain raw materials from Supplier 2
Raw materials acquired
Manufacture product
product not in stock
Check stock availability
Purchase order received
product in stock
Retrieve product from warehouse
Confirm order
Ship and invoice
Get shipment address
Ship product
Order confirmed
Emit invoice
Receive payment
Order shipped and invoiced
Archive order
Order fulfilled

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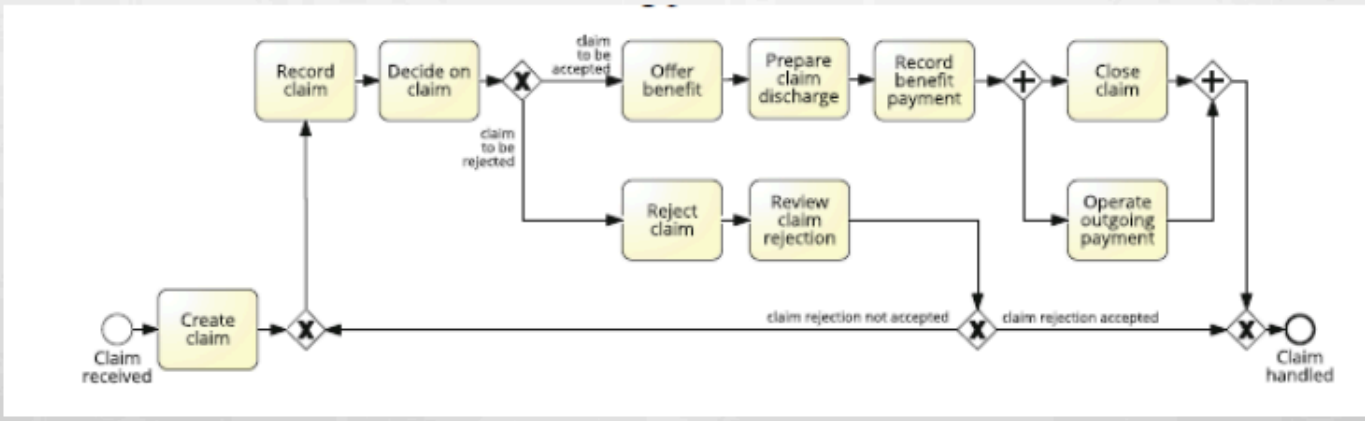

Acquire raw materials
Manufacture product
product not in stock
Check stock availability
Purchase order received
product in stock
Retrieve product from warehouse
Confirm order
Ship and invoice
Archive order
Order fulfilled

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Exercise 3.1 : describe the following model using one paragraph



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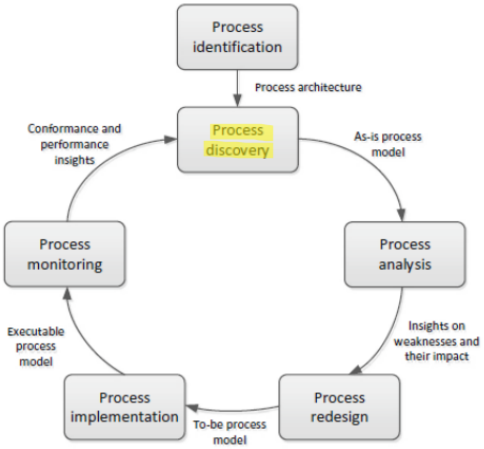
EXERCISE 3.2 : MODEL THE FOLLOWING BUSINESS PROCESS FOR HANDLING DOWNPAYMENTS

- The process for handling downpayments starts when a request for payment has been approved. It involves entering the downpayment request into the system, the automatic subsequent payment, the emission of the direct invoice and the clearance of the vendor line items. The clearing of the vendor line items can result in a debit or credit balance. In case of debit balance, the arrears are processed, otherwise the remaining balance is paid.

EXERCISE 3.3 : MODEL THE FOLLOWING BUSINESS PROCESS FOR UWU STUDENT ENROLMENT FOR CERTIFICATE COURSE

- University course enrollment: create a BPMN diagram for a university course enrollment process. Start with a student selecting courses and include tasks like verifying prerequisites, enrolling in courses, and issuing a confirmation. Incorporate decision gateways for scenarios where a course is full or prerequisites are not met.

Lecture 05 - Process Discovery



Process Discovery

Setting of Process Discovery

- It means collecting information about how a process currently works and creating an *as-is* process model.
- This stage is mainly about **gathering and organizing information**, not modeling yet.
- Modeling can begin **only after** enough information has been collected.
- First, we must create a good environment to gather information effectively.

Four main tasks

1. Define the setting
2. Gather information
3. Create the process model
4. Check the quality of the process model

Process Analysts vs Domain Experts

Process Analysts

- They collect information about how a business process works.
- They lead and guide the *process modeling* work.
- They work under the direction of the **process owner**.

Domain Experts

- They know the process deeply because they work in it every day.
- They understand how tasks are actually done in real life.
- They may be:
    - process participants
    - process owners
    - operational managers

External Roles

- People outside the organization who are connected to the process.
- Can include:
    - partners
    - suppliers
    - customers

Who to Hire as a Process Analyst

If you are hiring for a BPM project with an online bookstore:

- **Mike Miller**: 10 years experience with an online retailer, knows the order-to-cash process.
- **Sara Smith**: 5 years experience as a process analyst in banking, knows process modeling languages and tools.

Best Choice: *Sara Smith* – even though she hasn't worked in retail, she has the analytical skills, modeling knowledge, and BPM expertise needed to map and improve processes. Mike has domain knowledge, but lacks process modeling experience.

What Makes a Good Process Analyst:

- Picks the right people to involve.
- Tests hypotheses and finds patterns.
- Creates clear, readable process models.
- Improves processes from the *customer's point of view*.
- Explains BPM benefits like cost or time savings.
- Works with strategy and architecture teams.
- Good at running workshops.
- Always learning new skills.

Personality - Five factors:

- **Openness** – curious, appreciates ideas and planning.
- **Conscientiousness** (**හෘද සාක්ෂිය**) – disciplined, goal-oriented.
- **Extraversion** – positive, energetic, works well with others.
- **Agreeableness** – cooperative, compassionate.
- **Neuroticism** (**Emotional Stability**) – stays calm under pressure.

Challenges in Process Discovery:

- Knowledge is fragmented; needs repeated validation.
- The thinking process at the individual case level can be misleading.
- May not know modeling languages initially.

Example: A good analyst can take Mike's domain knowledge and combine it with process modeling skills to create an optimized order-to-cash process for the bookstore.

Process Discovery Methods

1. **Evidence-Based Discovery** – Collect information from real sources:
    - **Document analysis**: Study manuals, reports, or SOPs.
    - **Observation**: Watch how employees perform tasks.
    - **Automated process discovery**: Use software to track processes automatically.
      Example: Using system logs to see how orders are processed in an online store.
2. **Interview-Based Discovery** – Ask employees or managers about how tasks are done.
   Example: Talking to customer support staff to understand how complaints are handled.
3. **Workshop-Based Discovery** – Bring a group of stakeholders together to map and discuss processes.
   Example: A workshop with sales, warehouse, and delivery teams to map the order fulfillment process.

1.1 Evidence-Based Discovery – Document Analysis

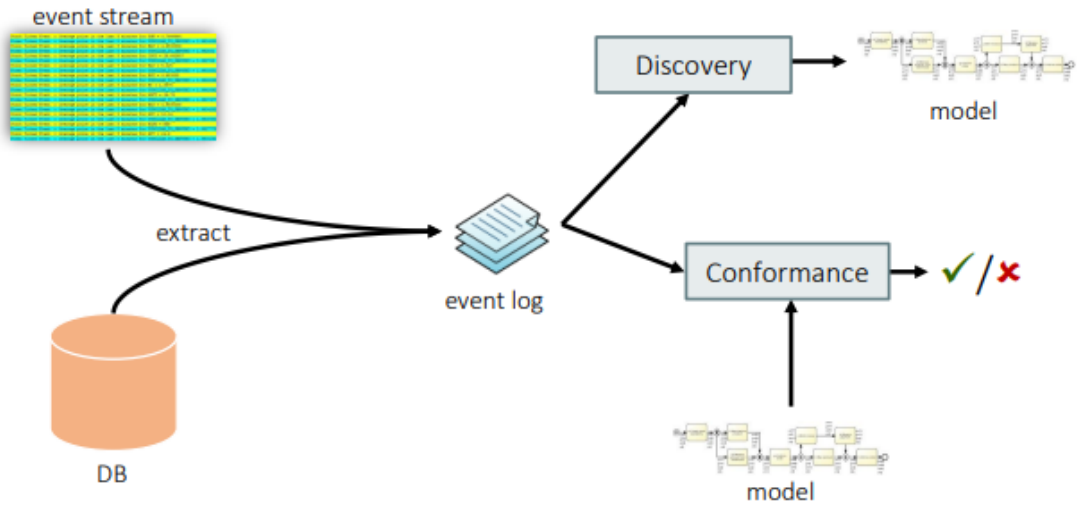
- Study documents to understand roles, tasks, and objects in a process: process descriptions, policies, org charts, work instructions, forms, quality reports, glossaries.
- Documents may not always be reliable or process-focused.
  Example: Reviewing an online store's SOPs to see how returns are handled.

1.2 Evidence-Based Discovery – Observation

- Watch how tasks are actually done:
    - **Active role:** Participate, e.g., act as a customer.
    - **Passive role:** Observe employees and their workflow.
- An active role gives hands-on experience but may miss the big picture.
- Passive role shows real behavior but may include participant bias.
  Example: Watching warehouse staff pack orders to see real workflow.

1.3 Evidence-Based Discovery – Automated Discovery / Process Mining

- Use **software to automatically extract process** data from event logs or databases.
- Helps build a process model and check if actual processes follow the rules.
  Example: Using an ERP system's logs to map the order-to-cash process automatically.



2.Interviews – Simplified



- Talk with stakeholders to gather process info and validate models.
- Types:
  - **forward** vs. **backward**,
  - **structured** vs. **unstructured**.
- Assumes both analyst and stakeholder understand the same terms.
- Risk: unusual or rare cases may be missed, so ask questions to capture exceptions.
  Example: Interviewing online store customer support staff to understand how refund requests are handled, including rare cases like partial refunds.

3.Workshops – Brown Paper Mapping



- Bring all key stakeholders together to map processes visually and create shared understanding.
- Usually one facilitator (process analyst), multiple domain experts, and sometimes the process owner.
- Can be supported by software, with a model created live, or done on large paper sheets ("brown-paper").
- Typically 3–5 half-day sessions.
  Example: Workshop with restaurant staff, delivery team, and app developers to map the full online food ordering process from order to delivery.

Discovery and Culture

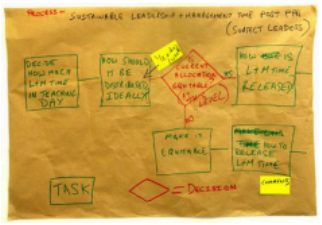
- Before starting the process discovery, understand the company's culture and mood.
- Open-culture companies encourage employees to share ideas and feedback freely; workshops work well here.
- In strict, hierarchical companies, ensure everyone has a chance to speak, and no ideas are held back.
  Example: A young tech startup may openly discuss workflow improvements in workshops, while a large hospital may need structured sessions to ensure all staff, including junior nurses, can share their ideas.

Strengths and Weaknesses

Technique	Strength	Weakness
Document Analysis	• Structured information • Independent from availability of stakeholders	• Outdated material • Wrong level of abstraction
Observation	• Context-rich insight into process	• Potentially intrusive • Stakeholders likely to behave differently • Only few cases
Automatic Discovery	• Extensive set of cases • Objective data	• Potential issue with data quality
Interview	• Detailed inquiry into process	• Requires sparse time of process stakeholders • Several iterations required before sign-off
Workshop	• Direct resolution of conflicting views	• Requires availability of several stakeholders at the same time

Organizing the Gathered Material

1. Identify the process boundaries
2. Identify activities and events
3. Identify resources and their handovers
4. Identify the control flow
5. Identify additional elements

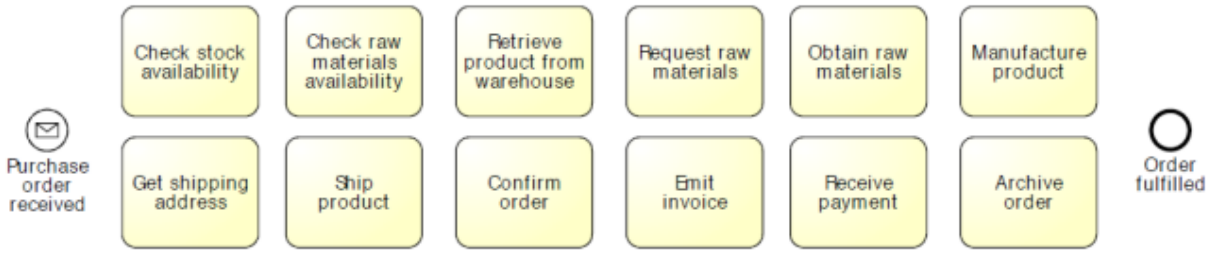


Organizing the Gathered Material

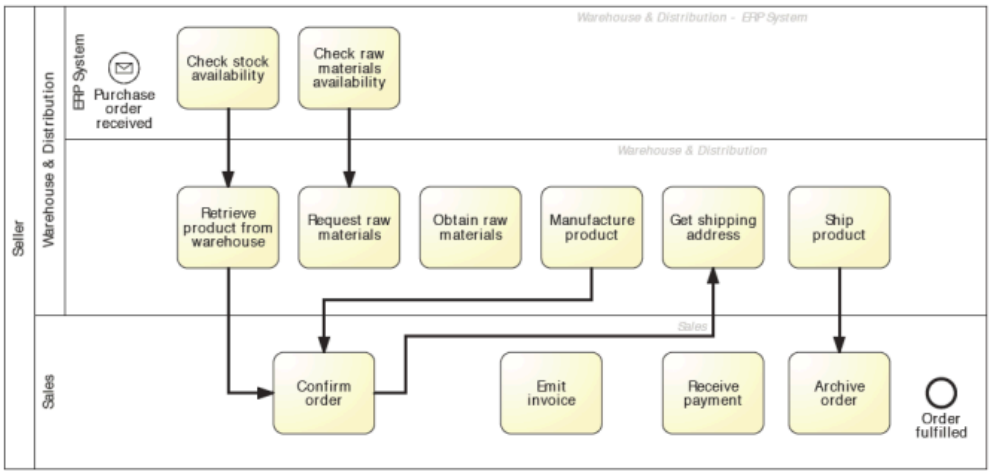
1. **Identify the process boundaries** – when the process starts and ends, what data is needed, and the perspective used.

- Under which condition does the process start?
- With which result does it end?
- Which perspective do we assume?
- What data are required as input and output to the process?

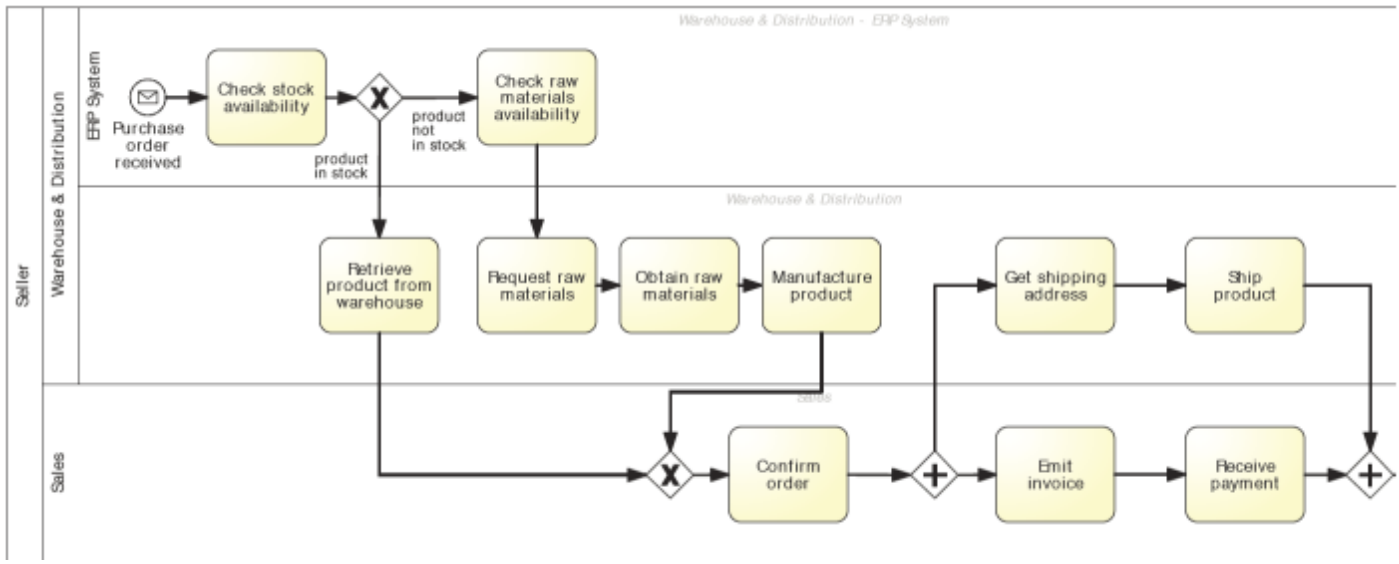
2. **Identify activities and events** – the tasks and triggers within the process.



3. **Identify resources and their handovers** – who or what is involved and how tasks are passed along.



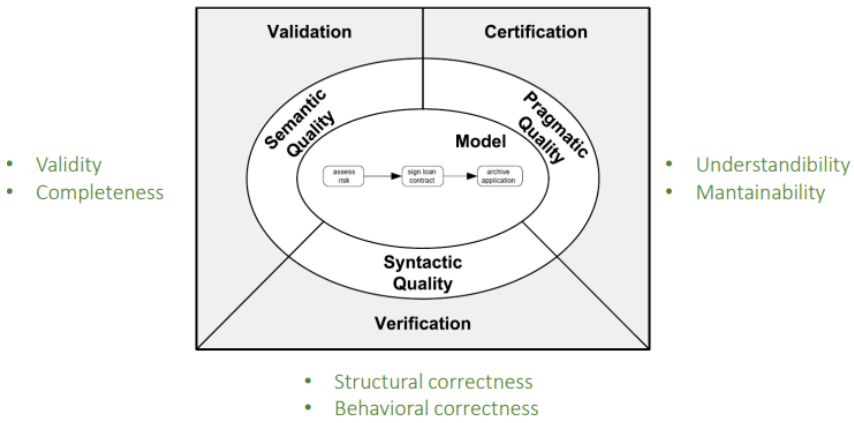
4. **Identify the control flow** – the sequence and rules of the process.



5. **Identify additional elements** – exceptions (rainy days) and other resources like data.

Example: In an online order process, the process starts when a customer places an order (boundary), activities include payment verification and packing (tasks), resources include staff and delivery system (handovers), the control flow defines the order sequence, and exceptions cover out-of-stock items or delayed shipments.

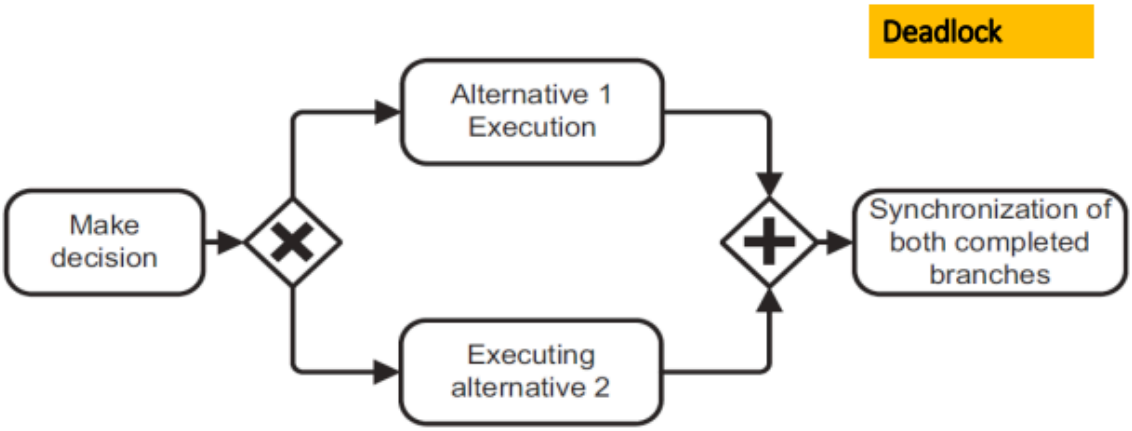
Process Modelling Quality Assurance



- **Validity** – The model accurately represents the real process.
- **Completeness** – All activities, events, and decisions are included.
- **Structural correctness** – The process flow is logically organized.
- **Behavioral correctness** – The process behaves as expected in different scenarios.
- **Understandability** – Easy for stakeholders to read and comprehend.
- **Maintainability** – Easy to update or change when needed.

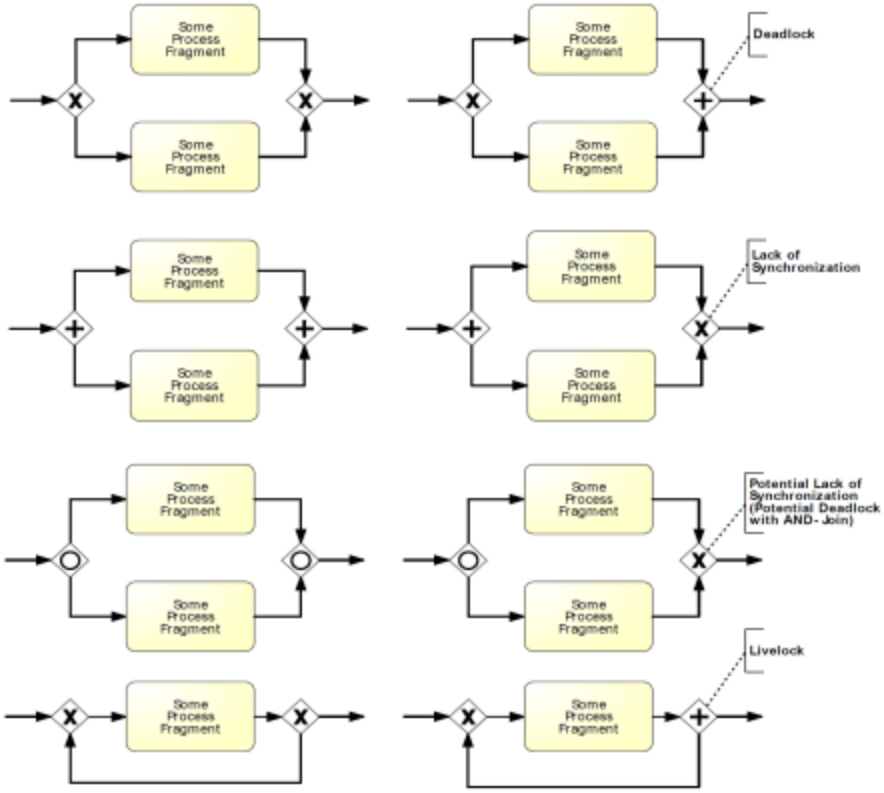
Example: For a food delivery app, the process model must include order placement, payment, cooking, delivery, and feedback (completeness), follow correct steps (structural correctness), handle late deliveries properly (behavioral correctness), be simple to read (understandability), and allow updates like adding new payment options (maintainability).

Is this process model of good quality?

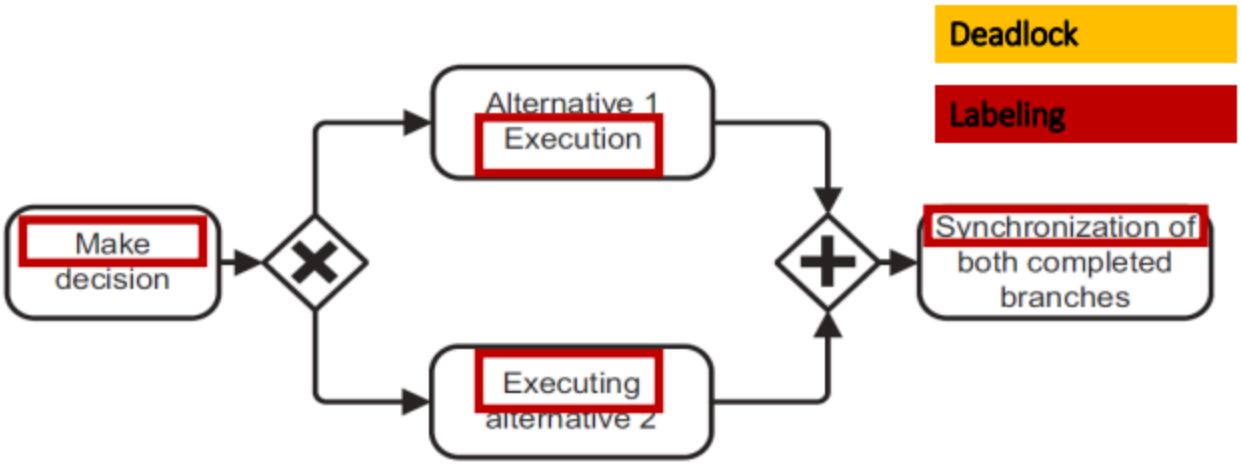


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Syntactic Quality: Verification



Is this process model of good quality?



Formulate Labels Adequately

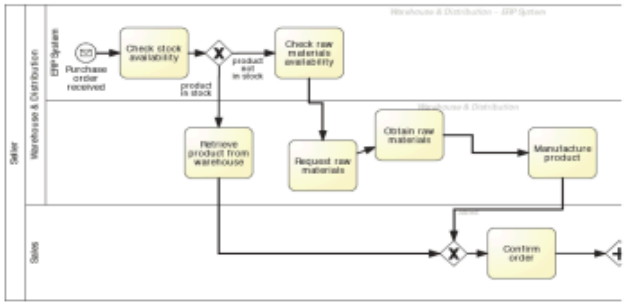
- **Activities** – Use **Verb-Object** format (e.g., *Approve Order*).
- **Events** – Use **Object-Past-Participle Verb** format (e.g., *Order Placed*).
- **Conditions** – Refer clearly to the related object (e.g., *If Payment Confirmed*).

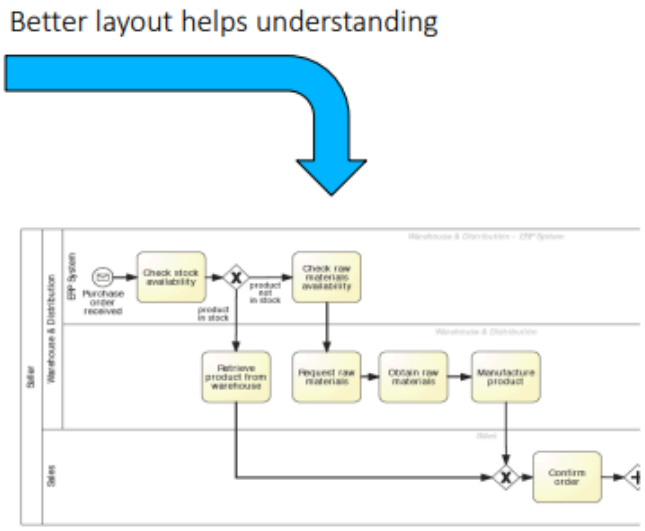
Semantic Quality:

- **Validity** – Labels accurately reflect real-world actions.
- **Completeness** – All relevant activities, events, and conditions are labeled.

Example: In an online bookstore, an activity could be *Pack Book*, an event could be *Book Shipped*, and a condition could be *If Stock Available*. Domain experts and process analysts validate that labels match reality and cover everything needed.

Pragmatic Quality – Example: Layout

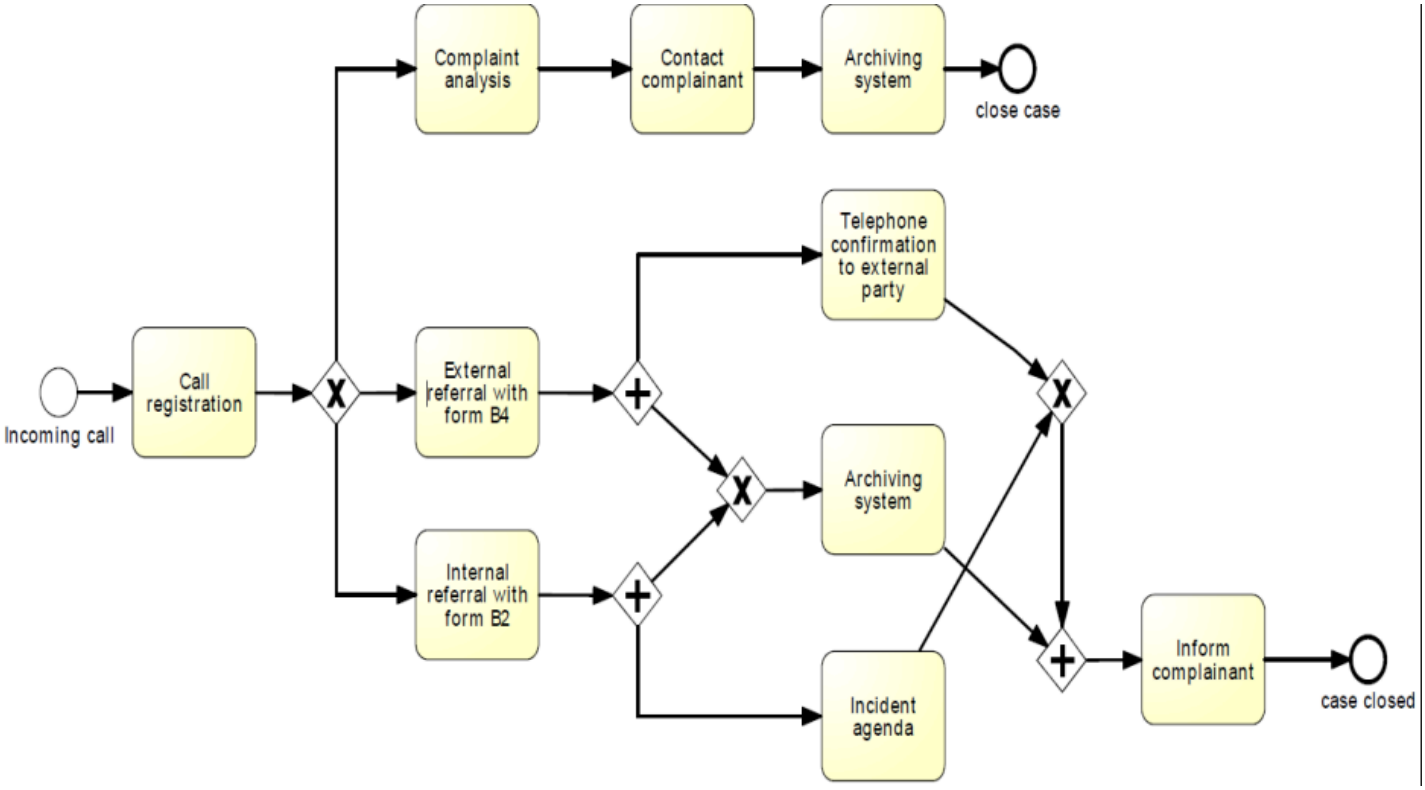




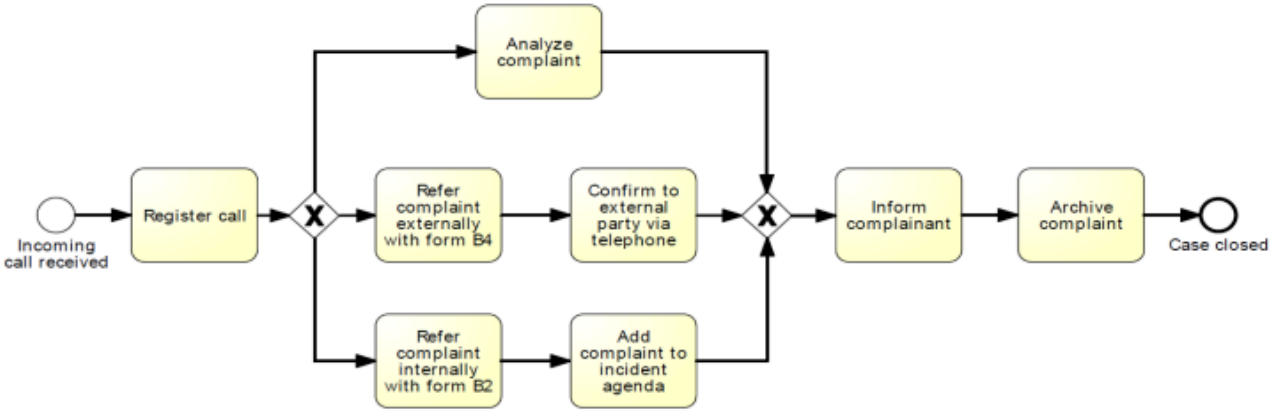
Seven Process Modeling Guidelines (7PMG)

G1 – Keep the model simple; use as few elements as possible.
G2 – Minimize the number of paths leading in or out of each element.
G3 – Use only one start event and one end event.
G4 – Make the model structured and organized.
G5 – Avoid OR-routing elements if possible.
G6 – Label activities using **Verb-Object** format (e.g., *Approve Order*).
G7 – Break down models with more than 30 elements into subprocesses.

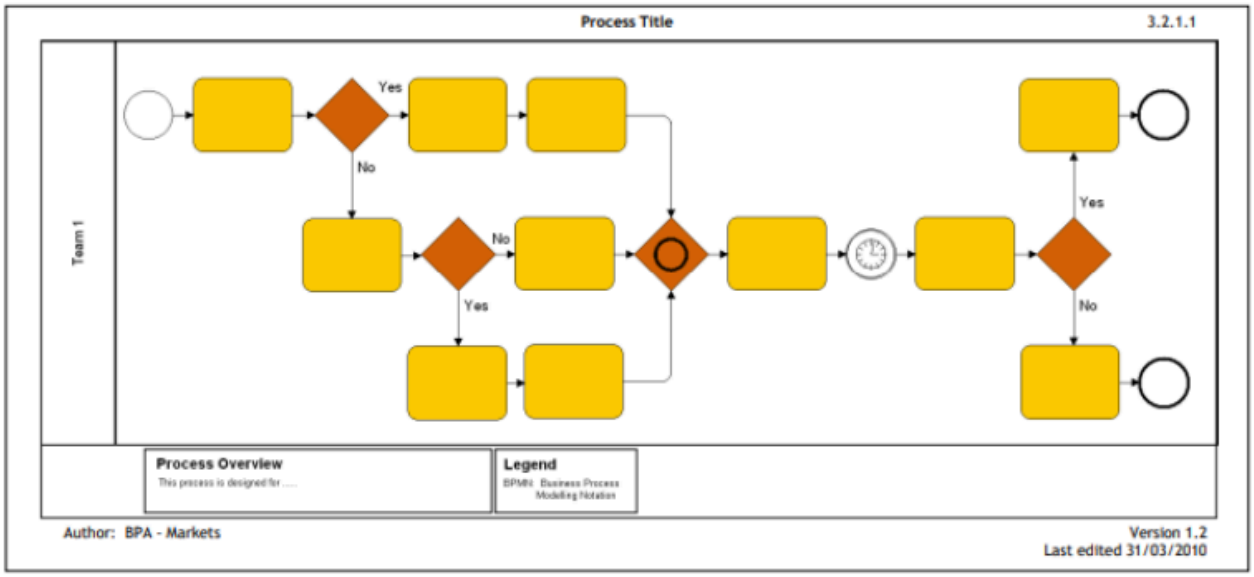
Example: In a food delivery process, instead of showing every minor step in one big model, split it into subprocesses like *Order Preparation* and *Order Delivery* to keep it clear and manageable.


Complaint analysis
Contact complainant
Archiving system
close case
Telephone confirmation to external party
Incoming call
Call registration
External referral with form B4
Archiving system
Internal referral with form B2
Inform complainant
case closed
Incident agenda

Solution


Analyze complaint
Incoming call received
Register call
Refer complaint externally with form B4
Confirm to external party via telephone
Inform complainant
Archive complaint
Case closed
Refer complaint internally with form B2
Add complaint to incident agenda

Lecture 6 - More about BPMN

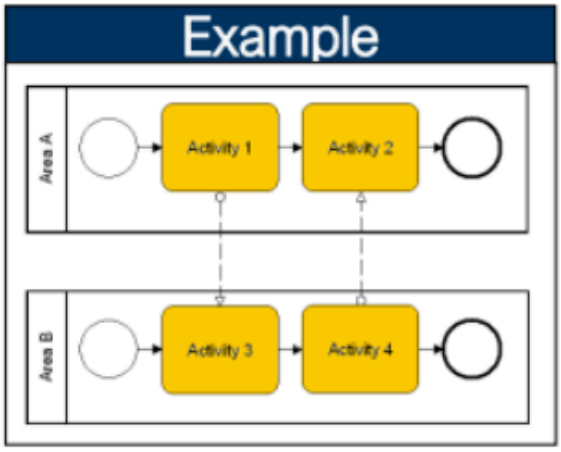


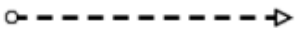
several events types

Trigger	Start	End	Description
General			The start event indicates where a particular process will start. The end event will indicate where the process will end.
Message			A message arrives from a participant and triggers the event. This causes the process to start or end.
Timer			A timer indicates a time based event that will trigger the start or continue the process. The timer can be used to model time based delays.
Link			A link is a mechanism for connecting the end of one process to the start of another

Message flows

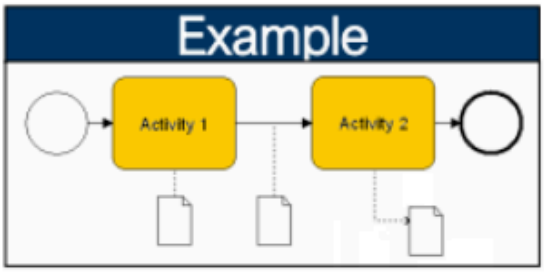
- A message flow is a dashed connector representing a signal (message) send between two pools. It may not be used to connect two nodes within a pool.

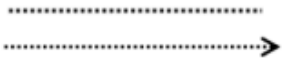




Associations indicate that a message or object (such as a document) is associated with that process

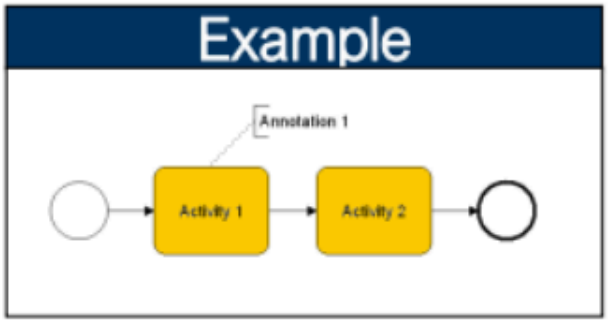
- In BPMN, a data object is linked either to a sequence flow, message flow, or an activity using a dotted line connector called an association.
- When lined to a sequence flow or message flow, the association is directional, with an arrow to or from the activity or event.

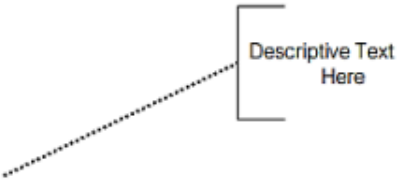




Text annotations

- Arbitrary text may be inserted in the diagram and linked to another object with a text annotation artifact and association connector.
- The text annotation is an enlarged left square bracket symbol





All processes must follow a verb-noun convention (i.echeck report)

Acquire	Borrow, buy, construct, develop, draw, lease, receive, rent
Activate	Authorise, enable, enact, establish, generate, guarantee, implement, initiate, install, integrate, open, reactivate, request
Administer	Admonish, apply, command, consult, direct, enhance, enquire, maintain, measure, monitor, operate, review, safeguard, supervise, waive
Decide	Accept, agree, allocate, approve, confirm, determine, identify, reject, select
Manipulate	Access, accumulate, add, archive, balance, calculate, compare, convert, create, delete, derive, exchange, file, modify, move, read, retrieve, sequence, store
Prepare	Analyse, define, draft, evaluate, forecast, gather, justify, negotiate, plan, propose, schedule, test, validate, verify
Provide	Acknowledge, announce, deliver, deposit, display, issue, lend, notify, offer, pay, print, report, return, send
Record	Capture, recognise, register
Suspend	Delay, make dormant
Terminate	Abandon, accomplish, cancel, close, complete, dispose of, sell, settle, withdraw

ster Class

Activity 1

A typical army recruitment process starts by shortlisting all candidates' applications. Those shortlisted are then called to take the following tests: drug and alcohol, eye, color vision, hearing, blood, urine, weight, fingerprinting, and doctor examination. The color vision can only be done after the eye test, while the doctor examination can only be done after color vision, hearing, blood, urine, and weight have been tested. Moreover, it may be required for some candidates to repeat some of these tests multiple times in order to get a correct assessment, e.g., the blood test may need to be repeated if the candidate has taken too much sugar in the previous 24 h. The candidates who pass all tests are asked to take a mental exam and a physical exam, followed by an interview. Only those who pass all these exams and perform well in the interview can be recruited in the army.

Activity 02 - Model the billing process of an Internet Service Provider (ISP).

The ISP sends an invoice by email to the customer on the first working day of each month (Day 1). On Day 7, the customer has the full outstanding amount automatically debited from its bank account. If an automatic transaction fails for any reason, the customer is notified on Day 8. On Day 9, the transaction that failed on Day 7 is re-attempted. If it fails again, on Day 10 a late fee is charged to the customer's bank account. At this stage, the automatic payment is no longer attempted. On Day 14, the Internet service is suspended until payment is received. If the payment is still outstanding on Day 30, the account is closed and a disconnection fee is applied. A debt-recovery procedure is then started.

Lecture 7 Process Analysis

Process Analysis

Analyzing

- Analyze the **as-is process** to identify opportunities for improvement.
- Activities to be done: Use **root cause analysis**, **value stream mapping**, or other techniques to assess inefficiencies.
- Evaluate the process against **performance metrics** and **business objectives**.
- Identify areas for **automation**, **streamlining**, or **redesign**.

Process Analysis Techniques

- **Value-Added & Waste Analysis**
  - Identifies steps that add value vs. steps that create waste.
- **Issue Register**
  - Records all issues, problems, and bottlenecks found in the process.
- **Root-Cause Analysis**
  - Finds the underlying cause of a problem (e.g., using the 5-Why method).
- **Flow Analysis**
  - Examines the flow of activities to detect delays or unnecessary steps.
- **Queuing Analysis**
  - Studies waiting times, queues, and workloads to reduce delays.
- **Simulation**
  - Models the process using software to test changes before implementation.

Value Added Activities

• Steps that directly produce value or satisfaction for the customer.
• Customers would be willing to pay for this step.
• Removing this step would reduce the value of the final output.

Example: *A bakery decorating a birthday cake — the customer pays for this final decorated product.*

Business Value-Added (BVA) Activities

• Steps needed for the business to operate smoothly or meet regulations.
• Helps collect revenue or support long-term business growth.
• Reduces business risks.
• Required to follow legal or industry rules.

Example: *A company preparing monthly financial statements — customers don't pay for it, but the business must do it for compliance.*

Waste Analysis

• Opposite of value-added analysis — identifies activities that add no value.

1.Move (Transportation + Motion)

- Wastes caused by unnecessary movement of people, materials, or equipment.

Example: *A worker walking back and forth because tools are stored far away.*

2.Hold (Inventory + Waiting)

- Wastes caused by storing items too long or waiting for the next step.

Example: *A machine stops because materials haven't arrived, causing waiting time.*

3.Overdo (Defects + Overprocessing + Overproduction)

- Wastes from doing more than required or producing too much.

Example:

- *Defect: A product with errors needing repair.*
- *Overprocessing: Adding features customers don't need.*
- *Overproduction: Making more items than customers ordered.*

Stakeholder Analysis and Issue Documentation

1. Stakeholder Analysis

• Identifies all individuals or groups affected by the process.
• Helps collect issues, needs, and expectations from multiple perspectives.
• Ensures that improvements consider customers, employees, managers, suppliers, etc.

Example: *In a hospital, doctors, nurses, patients, and admin staff each provide different issues about the patient admission process.*

2. Issue Register

• A structured document used to record problems found in the process.
• Captures issue description, impact, root cause, priority, and owner.
• Ensures no issue is forgotten or ignored during analysis.

Example: *"Long waiting time for approvals" is logged with its cause and priority for resolution.*

3. Pareto Analysis & PICK Charts

• Tools used to filter and prioritize issues for improvement.
• **Pareto Analysis (80/20 rule):** Identifies the few issues causing most problems.
• **PICK Chart:** Classifies issues as **P**ossible, **I**mplement, **C**hallenge, or **K**ill based on impact and effort.
• Helps teams focus on the most valuable improvements first.

Example: *Pareto analysis reveals that 20% of system errors cause 80% of delays, so these issues are prioritized for redesign.*

Stakeholders

Stakeholders are people or groups who are affected by a process or have an interest in how it performs. They fall into five categories:

1. Customers

- The people who receive the final product or service.
  Real-world example: A customer ordering food from an online delivery app.

2. Process Participants

- Employees who actually perform the work steps in the process.
  Real-world example: Warehouse staff picking and packing online orders.

3. External Parties

- Suppliers, partners, or subcontractors involved in the process.
  Real-world example: A delivery company transporting products for an e-commerce store.

4. Process Owner and Operational Managers

- People who supervise the process and ensure it runs properly.
  Real-world example: A production manager overseeing manufacturing operations.

5. Sponsor and Executive Managers

- Senior leaders who support and fund process improvement efforts.
  Real-world example: A company director approving a project to improve customer service speed.

Issues and Factors: Equipment Rental Process

• Clerk misunderstood the site engineer's requirements for the equipment.
• Clerk selected the wrong equipment from the supplier's catalog due to inattention.
• Clerk indicated an incorrect delivery date in the purchase order (PO).
• Supplier failed to deliver the ordered equipment.
• Delivered equipment is faulty or not ready for use.
• Supplier delivered the equipment to the wrong construction site or at the wrong time.
• Equipment arrived five working days after it was needed by the site engineer.
• Equipment was rejected by the site engineer.

Real-world example: *A construction site requests a crane for a project starting Monday. The clerk orders a smaller crane and sets the delivery for Thursday. The supplier delivers the wrong crane on Friday, which is not ready to use, so the site engineer rejects it, delaying the project.*

Issue 1: Equipment Kept Longer Than Needed

Priority: 1
Description: Site engineers keep the equipment longer than needed.
Data and Assumptions: BuildIT rents 3,000 pieces of equipment per year. In 10% of cases, site engineers keep equipment 2 extra days. Average rental cost = €100/day.
Qualitative Impact: Not applicable.
Quantitative Impact: 0.1 × 3,000 × 2 × 100 = €60,000 additional rental costs per year.

Real-world example: *A site requests a drill for 5 days but keeps it for 7 days. BuildIT charges extra for the two unused days, increasing costs.*

Issue 2: Rejected Equipment

Priority: 2
Description: Site engineers sometimes reject delivered equipment due to non-conformance

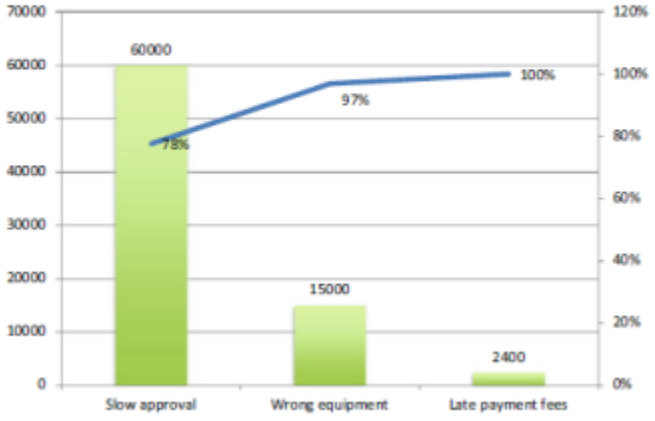
with specifications.
Data and Assumptions: BuildIT rents 3,000 pieces of equipment per year. 5% are rejected due to internal mistakes. Each rejection costs €100/day.
Qualitative Impact: Disrupts construction schedules, frustrates staff, and creates internal conflicts.
Quantitative Impact: 3,000 × 0.05 × 100 = €15,000 per year.

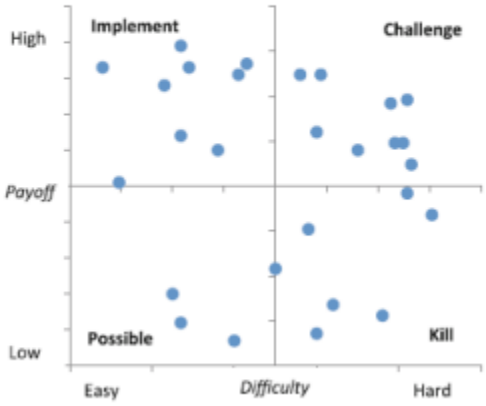
Real-world example: *A crane is delivered that doesn't meet weight requirements. The site engineer rejects it, causing schedule delays and extra cost.*

Issue Register Guidelines

• **Name of the issue:** Short, clear, understandable by all stakeholders (2–5 words).
• **Description:** 1–3 sentences focusing on the issue itself, not its consequences.
• **Priority:** Number indicating importance relative to other issues (e.g., 1 = highest).
• **Data and Assumptions:** Any data or rough estimates used to calculate impact (number of occurrences, estimated cost, etc.).
• **Qualitative Impact:** Effects on customer satisfaction, employee satisfaction, supplier relations, reputation, or other non-quantifiable impacts.
• **Quantitative Impact:** Estimated measurable effect like cost, time, or revenue loss. Links qualitative observations to numbers.

Data analysis

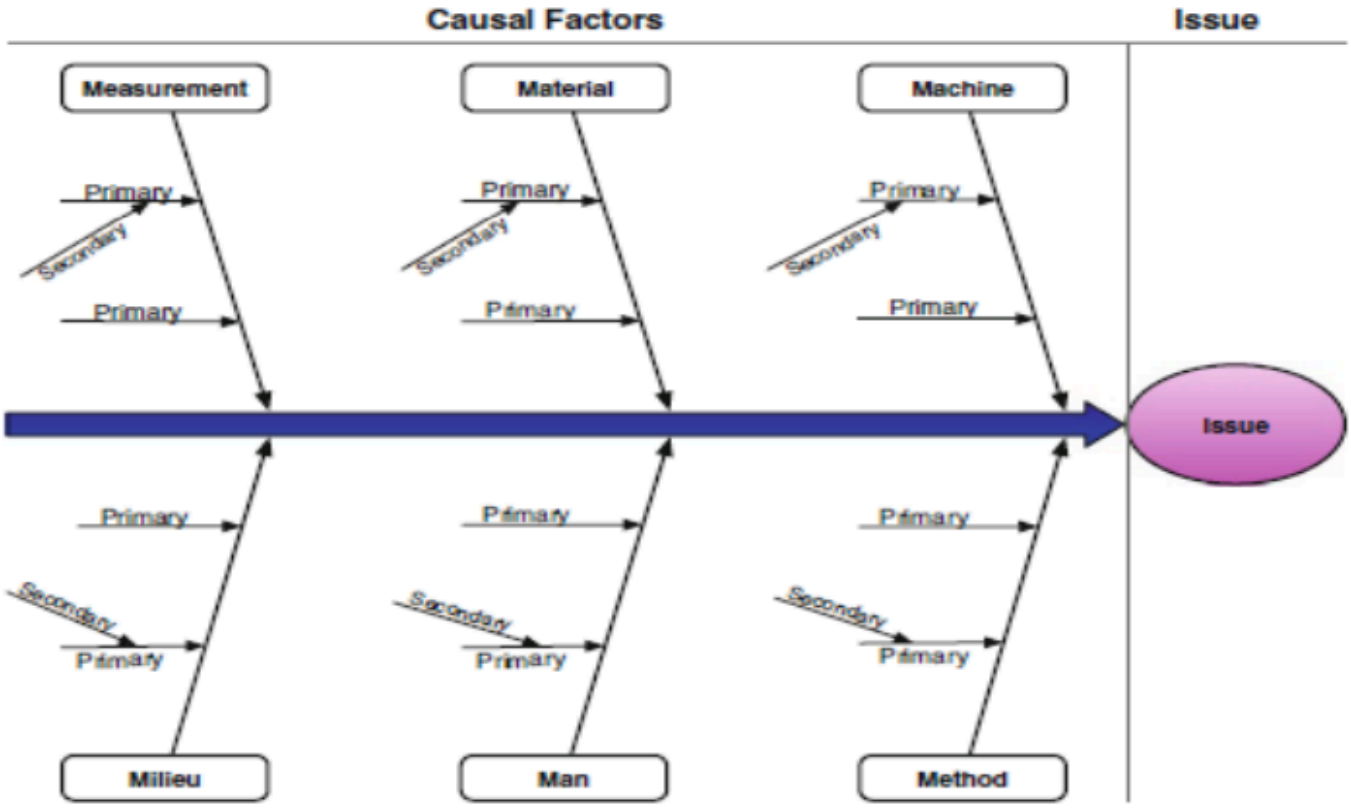




Root Cause Analysis – 6M's (Cause-Effect Analysis)

Root cause analysis helps identify **why a problem occurs** by examining six categories of possible causes, called the **6M's**:

1. **Machine** – Issues related to equipment or tools.
   Example: A printing machine jams frequently, causing delays.
2. **Method** – Problems in the process or procedure.
   Example: Workers follow outdated steps that slow down production.
3. **Material** – Defects or issues in raw materials or supplies.
   Example: Low-quality steel causes weak construction beams.
4. **Man** – Human errors or lack of skills.
   Example: An employee enters the wrong data in the system.
5. **Measurement** – Errors in inspection, testing, or monitoring.
   Example: Incorrect temperature readings cause baking errors.
6. **Milieu** – Environmental factors affecting the process.
   Example: High humidity damages electronic components during assembly.



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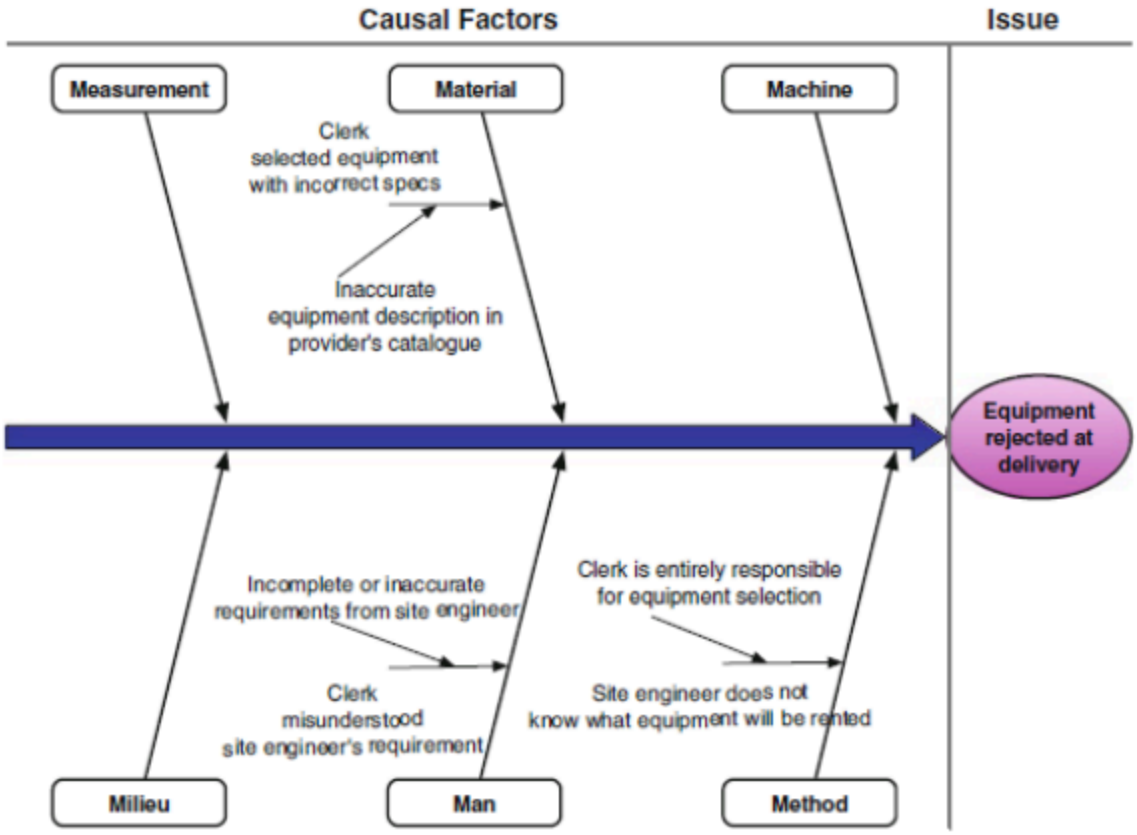
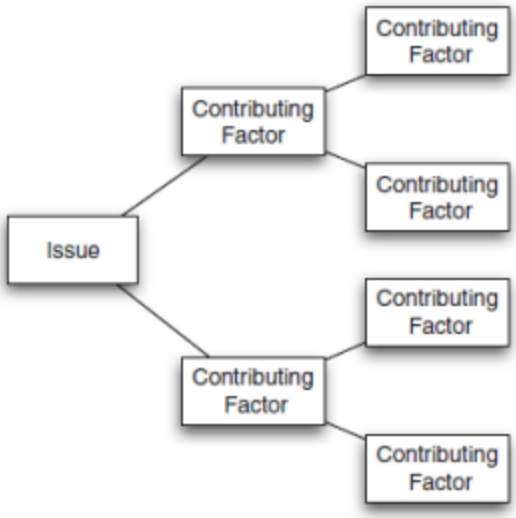


Fig. 6.6 Cause-effect diagram for issue "Equipment rejected at delivery"

Root Cause Analysis – Why-Why Diagrams



Root Cause Analysis – Why-Why Diagrams

• A problem is analyzed by repeatedly asking **"Why?"** until the root cause is identified.

Example Analysis:

- **Issue:** Site engineers sometimes reject delivered equipment.
    - **Why?** Wrong equipment is delivered.
        - **Why?** Miscommunication between site engineer and clerk.
            - **Why?** Site engineer provides incomplete or inaccurate specifications.
                - **Why?** Site engineer does not always see supplier catalogs and does not communicate with supplier.
                    - **Why?** Site engineer generally lacks Internet connectivity.
                - **Why?** Site engineer does not check the clerk's choice.
            - **Why?** Equipment descriptions in supplier catalogs are not accurate.

Real-world example: *A construction site rejects a crane because the clerk misread the order. The real reason is incomplete specifications from the engineer, compounded by poor catalog info.*

Questions and Answers

Q1: Explain Value-Added Analysis

• Identifies steps in a process that **directly create value for the customer**.
• Helps decide which steps are necessary, which are business value-added, and which are waste.
Example: Decorating a birthday cake is value-added; packaging it extra fancy may be waste.

Q2: What are the seven factors in Waste Analysis?

1. **Transportation** – Unnecessary movement of goods.
2. **Motion** – Unnecessary movement of people or equipment.
3. **Inventory** – Storing too much stock.
4. **Waiting** – Idle time between process steps.
5. **Defects** – Errors needing rework.
6. **Overprocessing** – Doing more than required.
7. **Overproduction** – Producing more than needed.
   Example: Making 200 items when only 120 are needed = overproduction waste.

Q3: Name five categories of stakeholders

1. Customers
2. Process participants
3. External parties (suppliers, subcontractors)
4. Process owner and operational managers
5. Sponsor and executive managers
   Example: In a bank, customers, tellers, IT vendors, branch manager, and director are stakeholders.

Q4: Describe Root Cause with 6M's

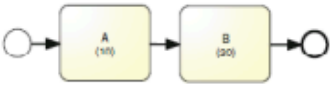
- **Machine:** Equipment issues.
- **Method:** Process or procedure problems.
- **Material:** Defective or poor-quality materials.
- **Man:** Human error or lack of skills.
- **Measurement:** Wrong monitoring, testing, or inspection.
- **Milieu:** Environmental factors affecting work.
  Example: A printing machine jams (Machine), because workers follow outdated steps (Method), and use poor paper quality (Material).

Q5: What is a Why-Why Diagram?

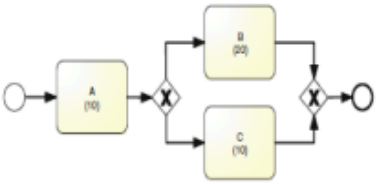
• A visual tool to find the **root cause of a problem** by asking "Why?" multiple times.
Example: Rejected construction equipment → wrong equipment → miscommunication → incomplete specifications → lack of internet → root cause.

Lecture 8 - Quantitative Process Analysis

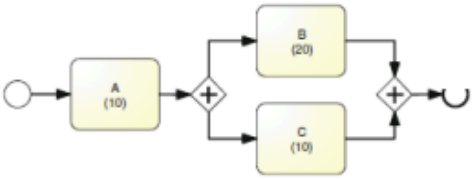
Flow Analysis - Calculating Cycle Time Using Flow Analysis



$$CT = \sum_{i=1}^{n} T_i$$



$$CT = \sum_{i=1}^{n} p_i \times T_i$$



$$CT = Max(T_1, T_2, \ldots, T_n)$$

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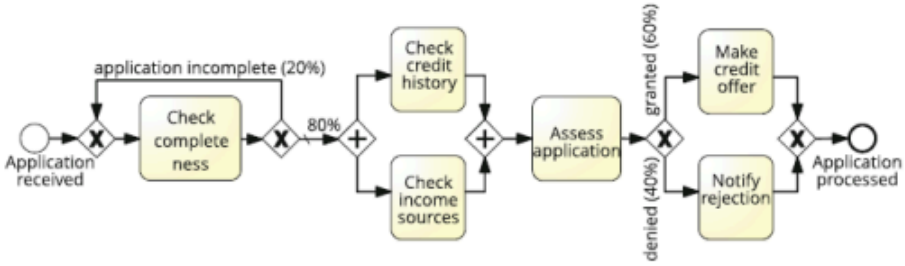
What is the average cycle time for the following scenario



$$CT = \frac{T}{1-r}.$$

.....

Example – Task 01 calculate the CT



Task Cycle	time
Check completeness	1 day
Check credit history	1 day
Check income sources	3 days
Assess application	3 days
Make credit offer	1 day
Notify rejection	2 days

Queue

• Queueing theory uses **mathematical techniques** to analyze systems where multiple cases compete for limited resources.
• Helps calculate important parameters such as:

- **Expected queue length** (how many cases wait)
- **Expected waiting time** (how long each case waits)

example: *Customers waiting at a bank counter — queueing theory predicts average wait times and number of customers in line.*

Simulation

• Process simulation generates many **hypothetical instances** of a process.
• Executes each step and records performance.
• Provides statistics like:

- Cycle times
- Average waiting times
- Average resource utilization

example: *A factory simulates production for 1,000 orders to find bottlenecks and optimize machine usage before real production.*

Question Answers

Q02: As a BPM analyst, what skills do you need to improve?
• Process mapping and modeling
• Data analysis and interpretation
• Problem-solving and root cause analysis
• Communication with stakeholders
• Knowledge of simulation and queueing techniques

Q03: Why is BPM important for organizations?
• Improves efficiency by reducing wasted time and resources
• Ensures process consistency
• Enhances customer satisfaction with faster, accurate service
• Allows organizations to adapt quickly to changes

Example: *An online retailer uses BPM to speed up order fulfillment and reduce delivery delays.*

Q04: What are the key elements of a business process?

1. **Inputs** – Resources required for the process
2. **Activities/Tasks** – Steps performed in the process
3. **Outputs** – Final product or service delivered
4. **Resources** – People, machines, or systems used
5. **Controls** – Policies, rules, or metrics ensuring process quality

Example: *In a loan approval process: application forms (input), verification steps (activities), approved loan (output), staff and software (resources), approval guidelines (controls).*

Q05: How would you prioritize which processes to improve first in a company?
• Use impact vs. effort analysis (focus on processes with **high impact, low effort**)
• Look at frequency and cost of errors or delays
• Consider customer satisfaction and regulatory importance

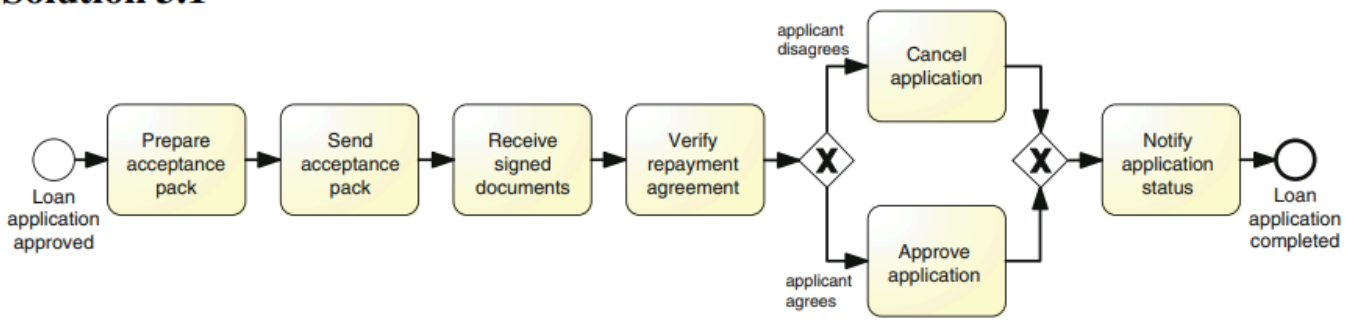
Example: *Fixing delays in customer order processing first because it affects revenue and satisfaction more than internal report filing.*

BPMN Questions

Exercise 3.1

Once a loan application has been approved by the loan provider, an acceptance pack is prepared and sent to the customer. The acceptance pack includes a repayment schedule which the customer needs to agree upon by sending the signed documents back to the loan provider. The latter then verifies the repayment agreement: if the applicant disagrees with the repayment schedule, the loan provider cancels the application; if the applicant agrees, the loan provider approves the application. In either case, the process completes with the loan provider notifying the applicant of the application status

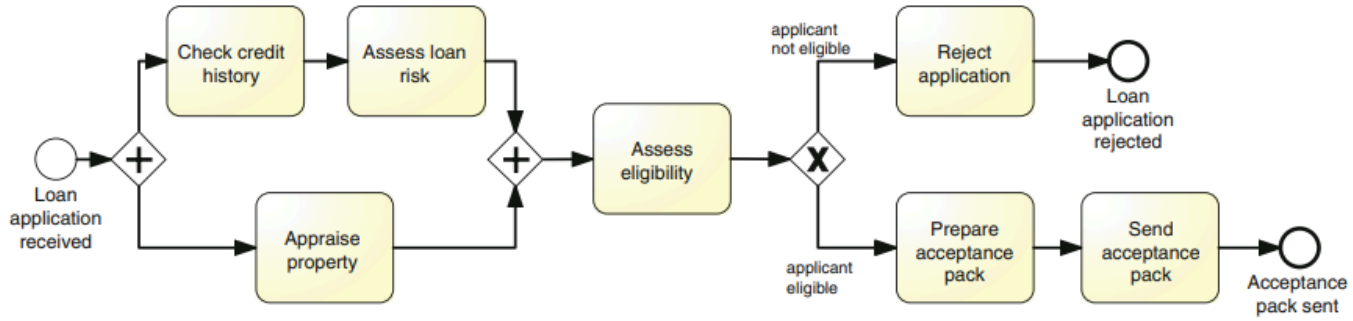
Solution 3.1



Exercise 3.2

A loan application is approved if it passes two checks: (i) the applicant's loan risk assessment, done automatically by a system, and (ii) the appraisal of the property for which the loan has been asked, carried out by a property appraiser. The risk assessment requires a credit history check on the applicant, which is performed by a financial officer. Once both the loan risk assessment and the property appraisal have been performed, a loan officer can assess the applicant's eligibility. If the applicant is not eligible, the application is rejected, otherwise the acceptance pack is prepared and sent to the applicant

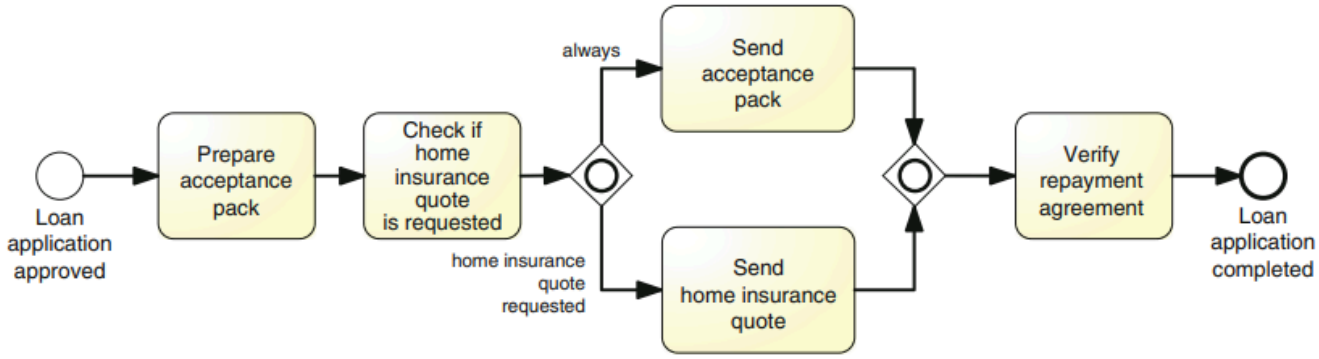
Solution 3.2



Exercise 3.3

A loan application may be coupled with a home insurance which is offered at discounted prices. The applicants may express their interest in a home insurance plan at the time of submitting their loan application to the loan provider. Based on this information, if the loan application is approved, the loan provider may either only send an acceptance pack to the applicant, or also send a home insurance quote. The process then continues with the verification of the repayment agreement

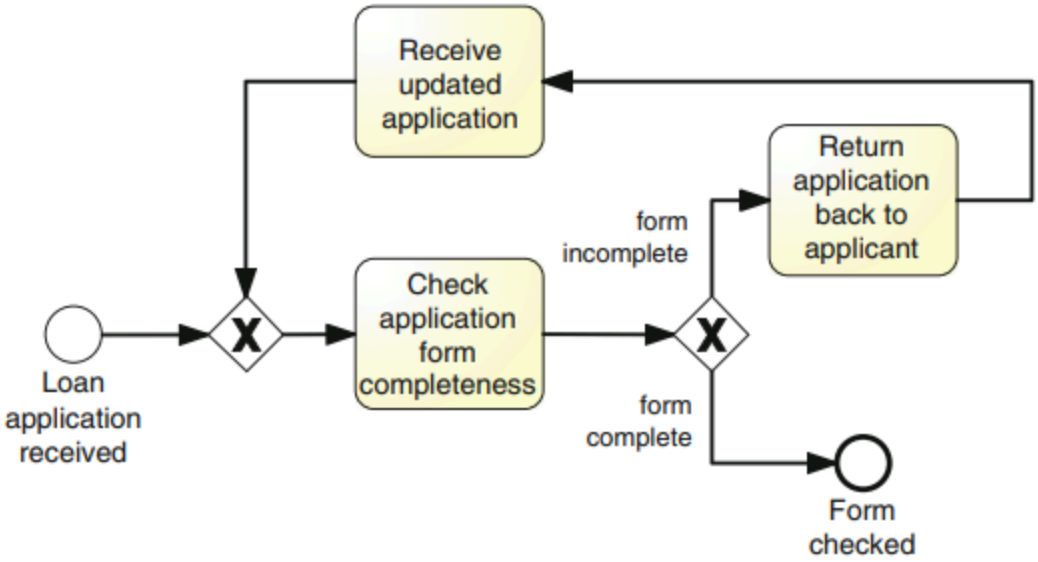
Solution 3.3



Exercise 3.4

Once a loan application is received by the loan provider, and before proceeding with its assessment, the application itself needs to be checked for completeness. If the application is incomplete, it is returned to the applicant, so that they can fill out the missing information and send it back to the loan provider. This process is repeated until the application is found complete.

Solution 3.4



Exercise 3.5

Is there any missing data object or data store in the example of Figure 3.14 (page 94)?

Question

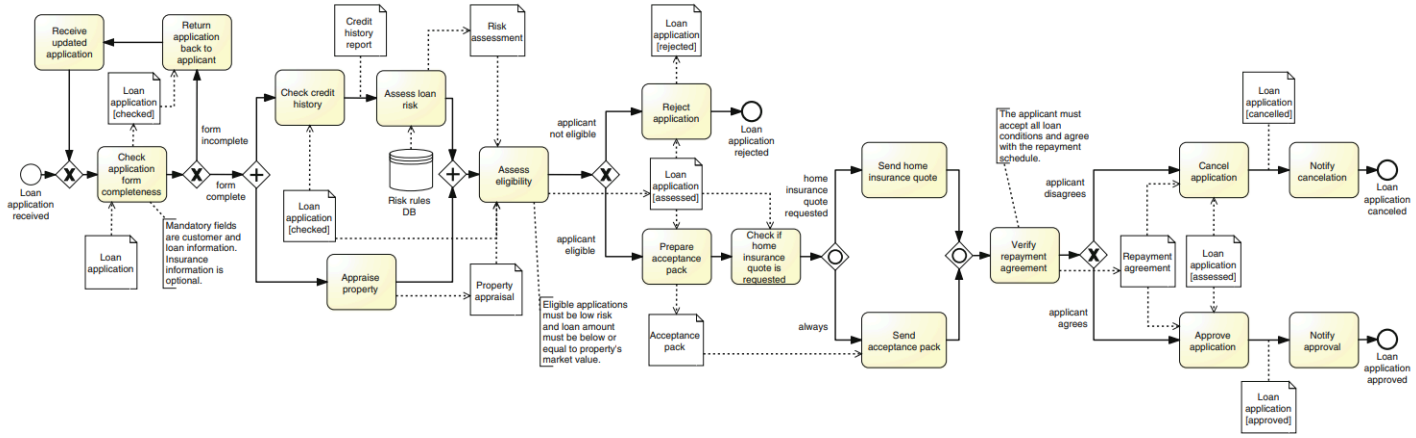
Do we always need to model data objects?

Data objects help the reader understand the flow of information and material from one activity to the other. However, the price to pay is an increased complexity of the diagram. Thus, we suggest using them only when they are needed for a specific purpose, for example when we later want to use the process model to communicate with an IT application development team in order to automate the process (cf. Chapter 10). There are cases in which we may need to provide additional information to the process model reader, for the sake of improving the understanding of the model. For example, in the order-to-cash process we may want to specify that activity “Ship product” includes the packaging of the product. Also, we may want to clarify

Solution 3.5 Activities “Retrieve product from warehouse” and “Manufacture product” require the purchase order as input, to identify which product to take from the warehouse or to build. Likewise, activities “Get shipment address” and “Emit invoice” require the confirmed purchase order as input, while activity “Receive payment” requires the payment as input, besides the confirmed purchase order. Anything else?

Exercise 3.6

Put together the four fragments of the loan assessment process that you created in Exercises 3.1–3.4.
Hint. Look at the labels of the start and end events to understand the order dependencies among the various fragments. Then extend the resulting model by adding all the required business objects. Moreover, attach annotations to specify the business rules behind (i) checking an application completeness, (ii) assessing an application eligibility, and (iii) verifying a repayment agreement



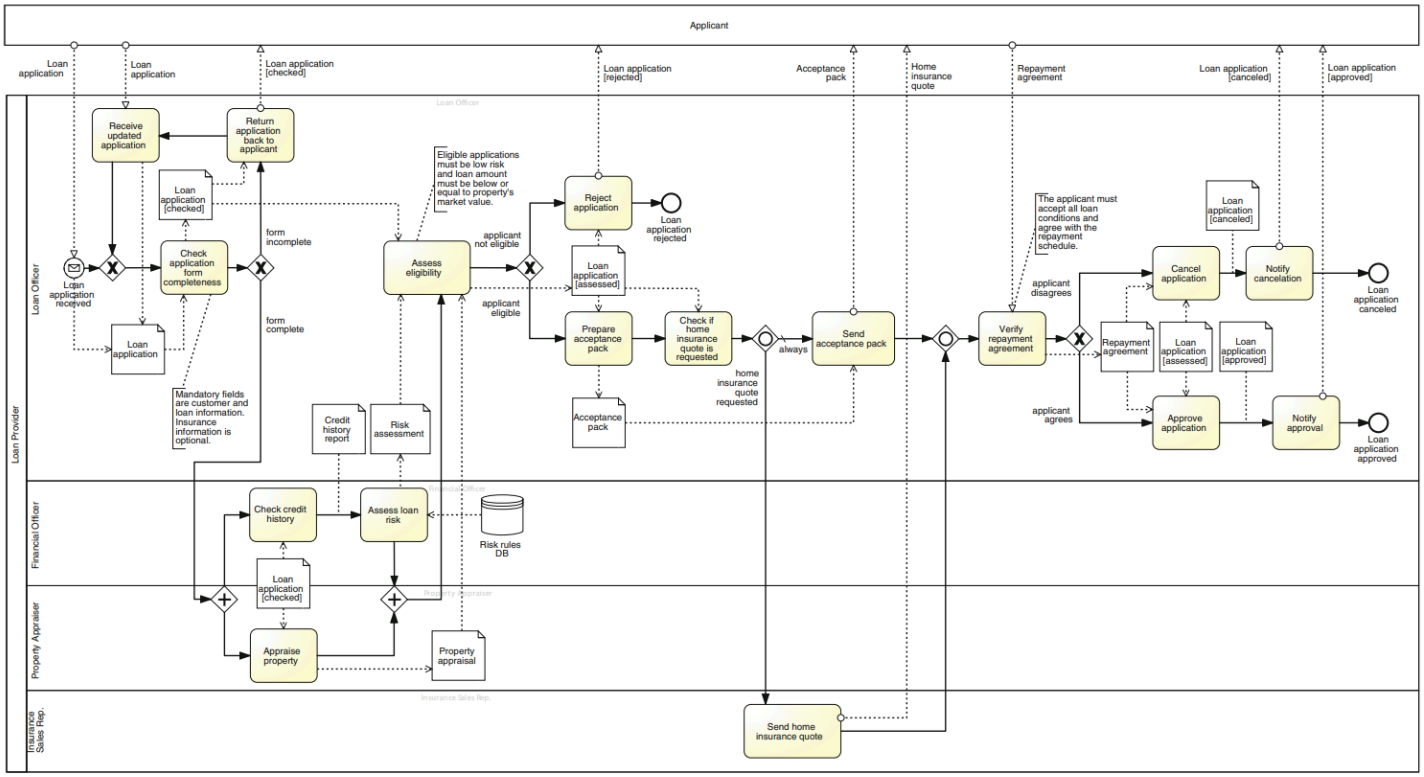
Exercise 3.7
Extend the business process for assessing loan applications that you created in Exercise 3.6 on page 96 by considering the following resource aspects.
The process for assessing loan applications is executed by four roles within the loan provider: a financial officer takes care of checking the applicant's credit history; a property appraiser is responsible for appraising the property; an insurance sales representative sends the home insurance quote to the applicant if this is required. All other activities are performed by the loan officer who is the main point of contact with the applicant.

Solution 3.7 See the Loan Provider pool in the model of Solution 3.8.

Exercise 3.8

Extend the model of Exercise 3.7 by representing the interactions between the loan provider and the applicant.

In the order-to-cash example we used pools to represent business parties and lanes to represent the departments and systems within the sales organization. This is because we wanted to focus on the interactions between the seller, the customer and the two suppliers. As mentioned before, this is the typical use for pools and lanes. However,since BPMN does not prescribe what specific resource types should be associated with pools and lanes, we may use these elements differently. For example, if the focus is on the interactions between the departments of an organization, we can model each department with a pool, and use lanes to partition the departments,e.g. in units or roles. In any case, we should avoid using pools and lanes to capture participants by their names since individuals tend to change frequently within an organization; rather, we should use the participant's role, e.g. financial officer.On the otherhand,we can use pools and lanes to represent a software system, e.g. an ERP system, since such systems are used for long periods of time.

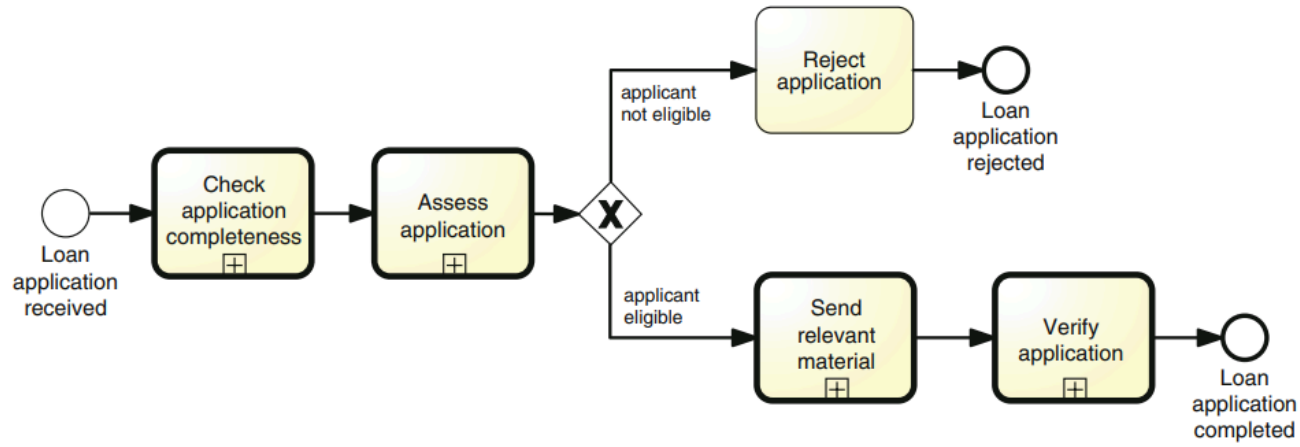


Exercise 3.9
Identify suitable sub-processes in the process for assessing loan applications modeled in Exercise 3.6 (page 96).
Hint. Use the building blocks that you created throughout Exercises 3.1–3.4.

Collapsing a sub-process does not imply losing its content. The sub-process is still there, just defined at an abstraction level below. In fact, we can nest sub processes in multiple levels, so as to decompose a process model hierarchically. An example is shown in Figure 3.19, which models a business process for disbursing home loans. In the first level we identified two sub-processes: one for checking the applicant's liability, the other for signing the loan. In the second level, we factored out the scheduling of the loan disbursement within the process for signing loans into a separate sub-process.

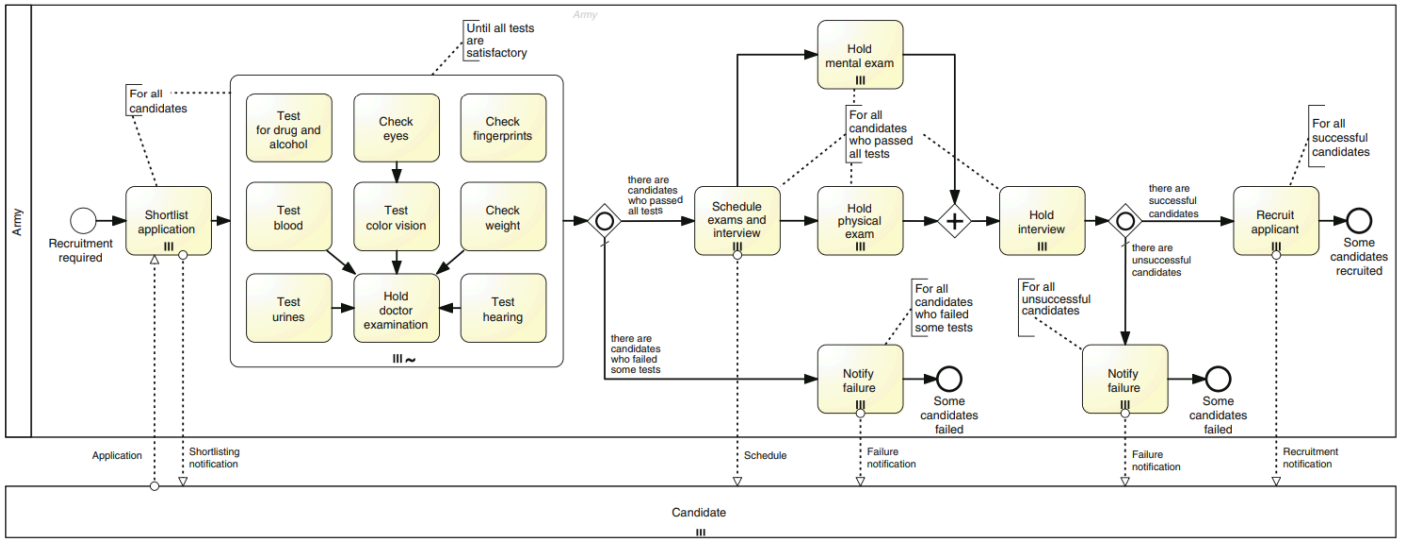
As we go down the hierarchical decomposition of a process model, we can add more details. For example, we may establish a convention that at the top level we only model core business activities, at the second level we add decision points, and so on all the way down to modeling exceptions and details that are only relevant for process automation.

Solution 3.9



Exercise 4.3
Model the following process snippet. A typical army recruitment process starts by shortlisting all candidates' applications. Those shortlisted are then called to take the following tests: drug and alcohol, eye, color vision, hearing, blood, urine, weight, fingerprinting, and doctor examination. The color vision can only be done after the eye test, while the doctor examination can only be done after color vision, hearing, blood,

urine, and weight have been tested. Moreover, it may be required for some candidates to repeat some of these tests multiple times in order to get a correct assessment, e.g., the blood test may need to be repeated if the candidate has taken too much sugar in the previous 24 h. The candidates who pass all tests are asked to take a mental exam and a physical exam, followed by an interview. Only those who pass all these exams and perform well in the interview can be recruited in the army



Exercise 4.4

Is there any other activity in the loan assessment model of Solution 3.8 (page 111) that can be replaced by a message event?

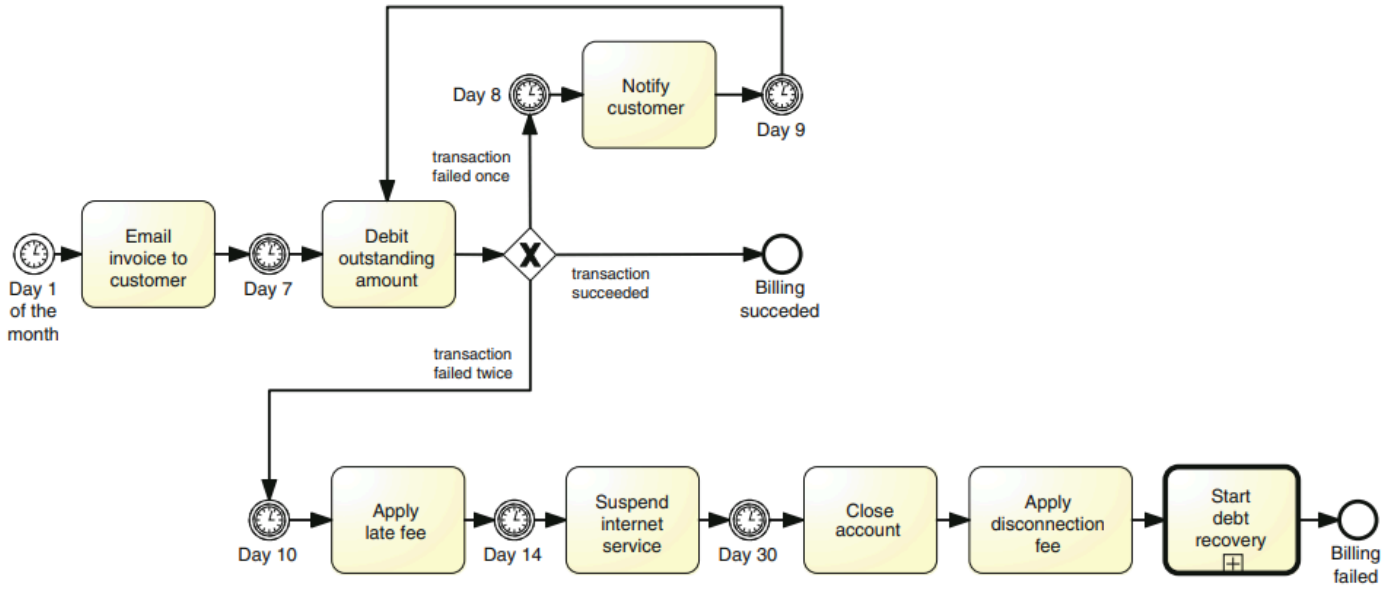
In BPMN, events come in two flavors based on the filling of their marker. A marker with no fill, like that on the start message event, denotes a catching event, i.e., an event that catches a trigger, typically originating from outside the process. A marker with a dark fill like that on the end message event denotes a throwing event, i.e., an event that throws a trigger from within the process. An intermediate message event can be used in both as a catching event (the message is received from another pool) or as a throwing event (the message is sent to another pool)

Solution 4.4 The activity "Send acceptance pack" can be replaced by an intermediate send message event; activities "Notify cancelation" and "Notify approval" can each be replaced by an end message event, thus removing the last XOR-join and the untyped end event altogether. Note that the activity "Send home insurance quote" cannot be replaced by a message event since it subsumes the preparation of the quote. In fact, a more appropriate label for this activity would be "Prepare home insurance quote". Similarly, we cannot get rid of activity "Reject application" as this activity changes the status of the application before sending the latter out.

Exercise 4.5 Model the billing process of an Internet Service Provider (ISP).

The ISP sends an invoice by email to the customer on the first working day of each month (Day 1). On Day 7, the customer has the full outstanding amount automatically debited from its bank account. If an automatic transaction fails for any reason, the customer is notified on Day 8. On Day 9, the transaction that failed on Day 7 is re-attempted. If it fails again, on Day 10 a late fee is charged to the customer's bank account. At this stage, the automatic payment is no longer attempted. On Day 14, the Internet service is suspended until payment is received. If the payment is still outstanding on Day 30, the account is closed and a disconnection fee is applied. A debt-recovery procedure is then started.

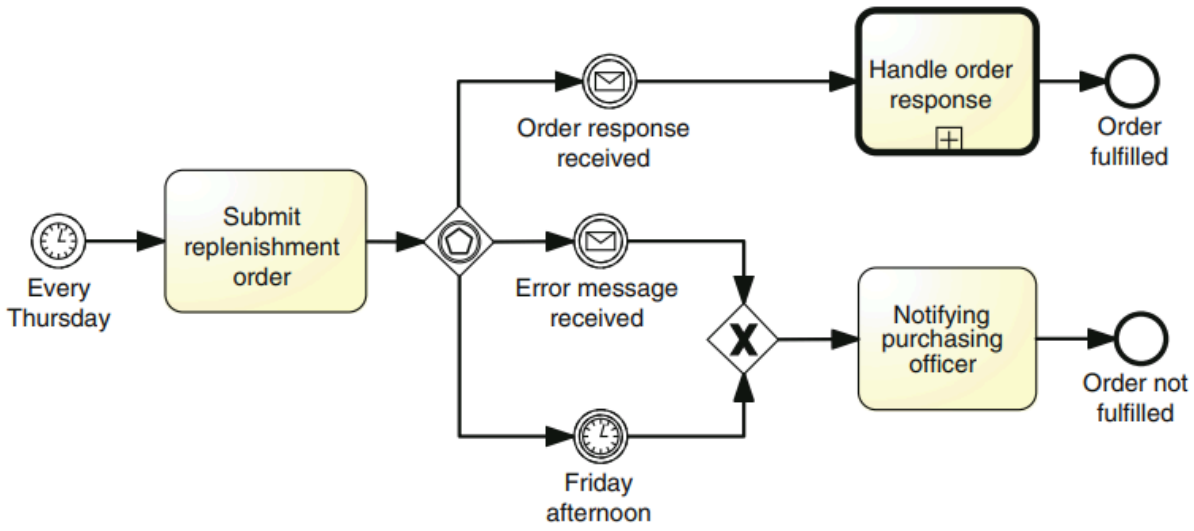
Solution 4.5



Exercise 4.6 Model the following process.

A restaurant chain submits a purchase order (PO) to replenish its warehouses every Thursday. The restaurant chain's procurement system assumes the receipt of either a "PO Response" or an error message. However, it may also happen that no response is received at all due to system errors or due to delays in handling the PO on the supplier's side. If no response is received by Friday afternoon or if an error message is received, a purchasing officer at the restaurant chain's headquarters should be notified. Otherwise, the PO Response is processed normally.

Solution 4.6


Every Thursday
Submit replenishment order
Order response received
Error message received
Friday afternoon
Handle order response
Order fulfilled
Notifying purchasing officer
Order not fulfilled

Exercise 4.7

Fix the collaboration diagram in Figure 4.13.

Acknowledgment This exercise is partly inspired by [92].

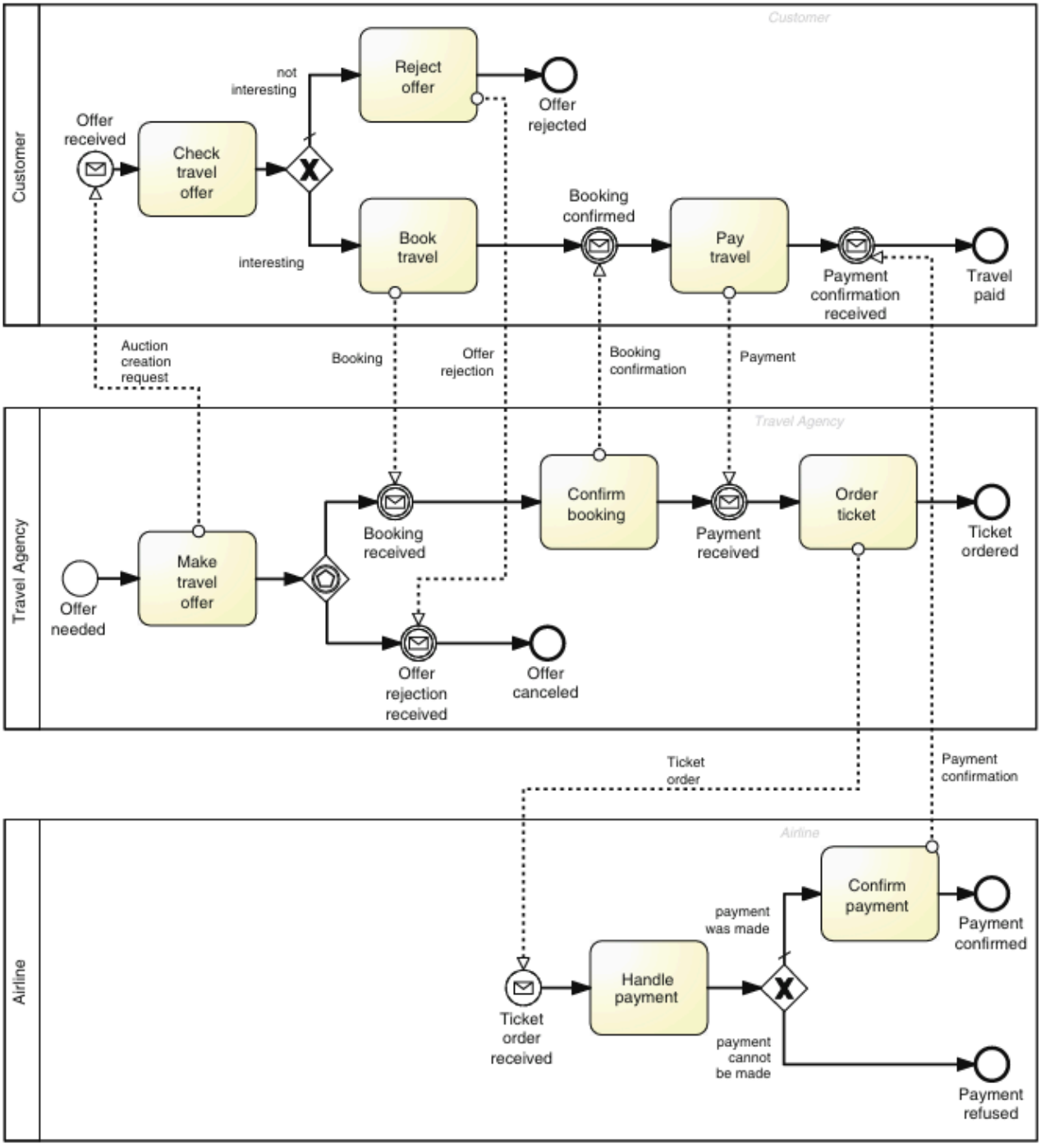
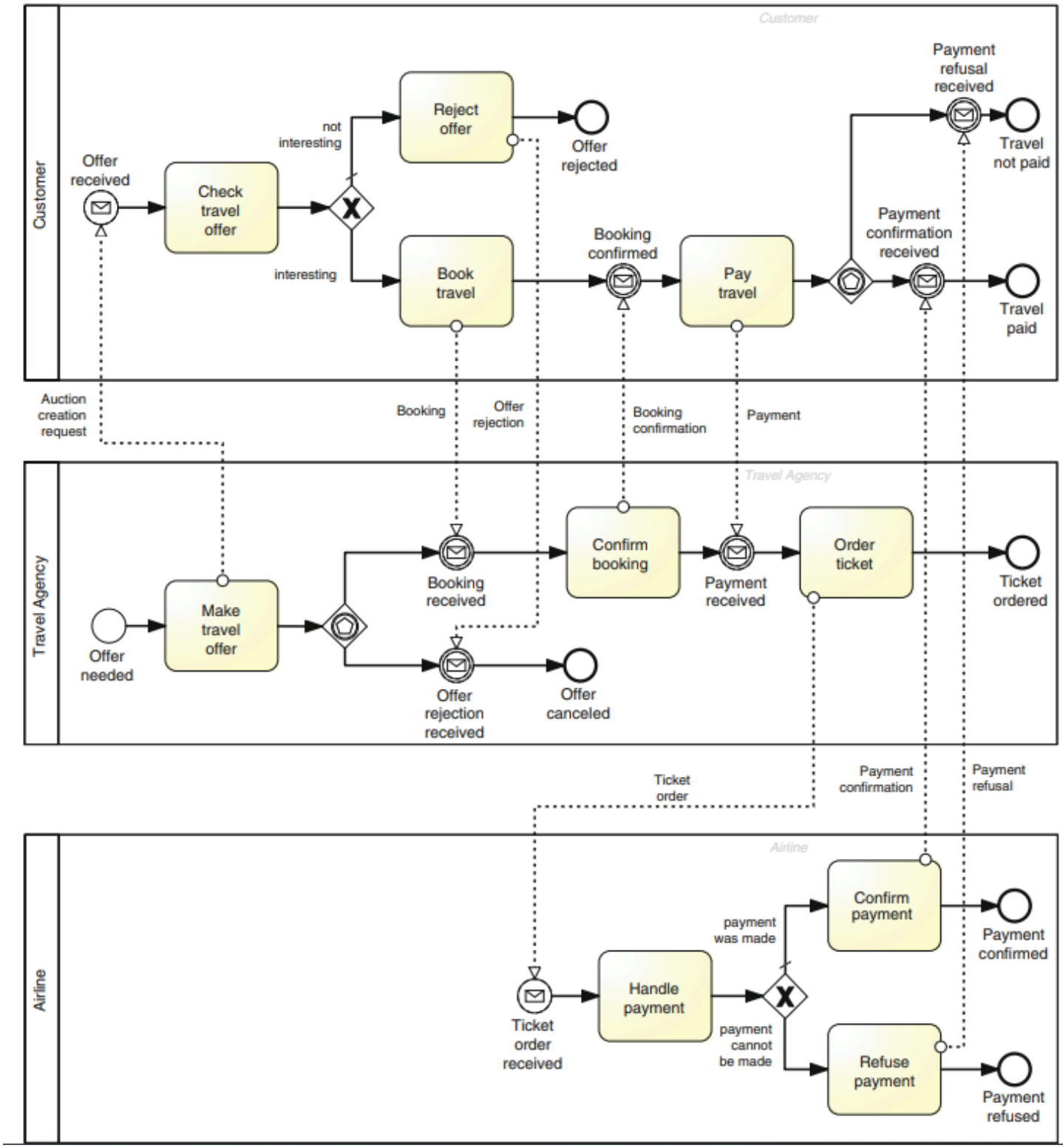


Fig. 4.13 A collaboration diagram between a customer, a travel agency, and an airline

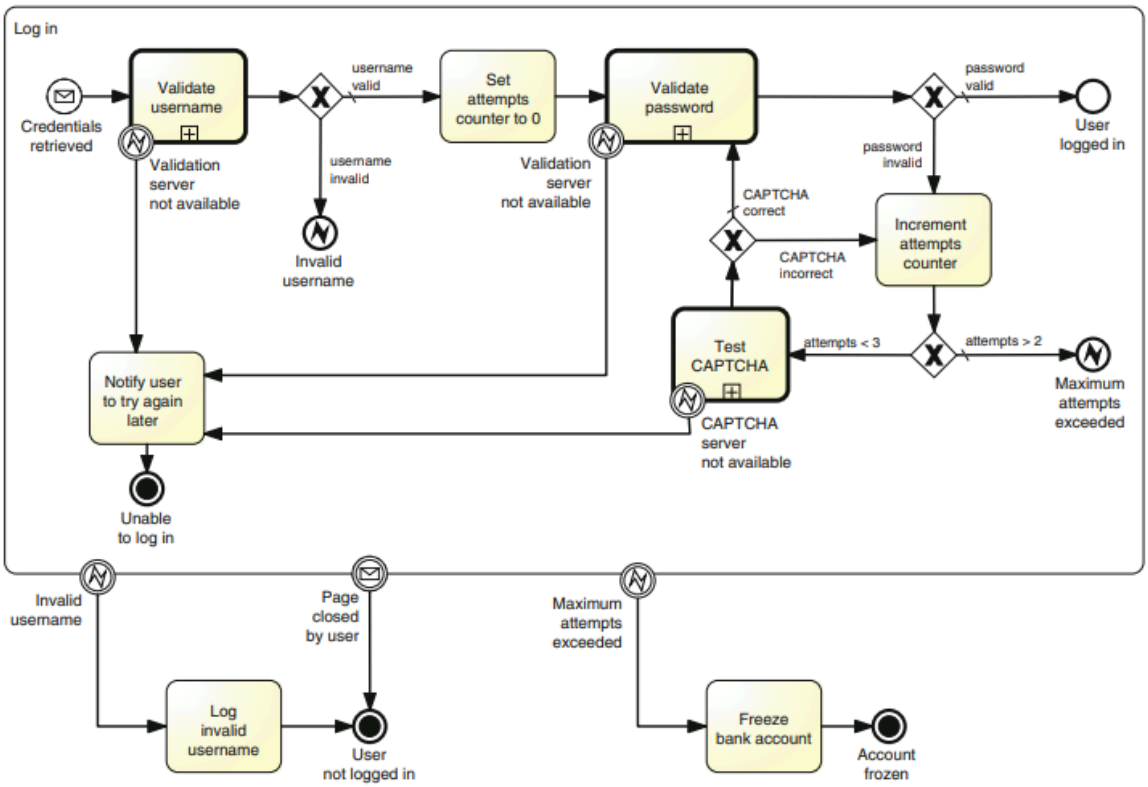
Solution 4.7


Customer
Offer received
Check travel offer
not interesting
Reject offer
Offer rejected
interesting
Book travel
Booking confirmed
Pay travel
Payment refusal received
Travel not paid
Payment confirmation received
Travel paid
Auction creation request
Booking
Offer rejection
Booking confirmation
Payment
Travel Agency
Offer needed
Make travel offer
Booking received
Confirm booking
Payment received
Order ticket
Ticket ordered
Offer rejection received
Offer canceled
Ticket order
Payment confirmation
Payment refusal
Airline
Ticket order received
Handle payment
payment was made
payment cannot be made
Confirm payment
Payment confirmed
Refuse payment
Payment refused

Exercise 4.9
Model the following routine for accessing an Internet bank service.

The routine for logging into an Internet bank account starts once the credentials entered from the user have been retrieved. First, the username is validated. If the username is not valid, the routine is interrupted and the invalid username is logged. If the username is valid, the number of password trials is set to zero. Then, the password is validated. If this is not valid, the counter for the number of trials is incremented and if lower than three, the user is asked to enter the password again, this time together with a CAPTCHA test to increase the security level. If the number of failed attempts reaches three times, the routine is interrupted and the account is frozen. Moreover, the username and password validation may be interrupted should the validation server not be available. Similarly, the server to test the CAPTCHA may not be available at the time of log in. In these cases, the procedure is interrupted after notifying the user to try again later. At any time during the log in routine, the customer may close the web page, resulting in the interruption of the routine.

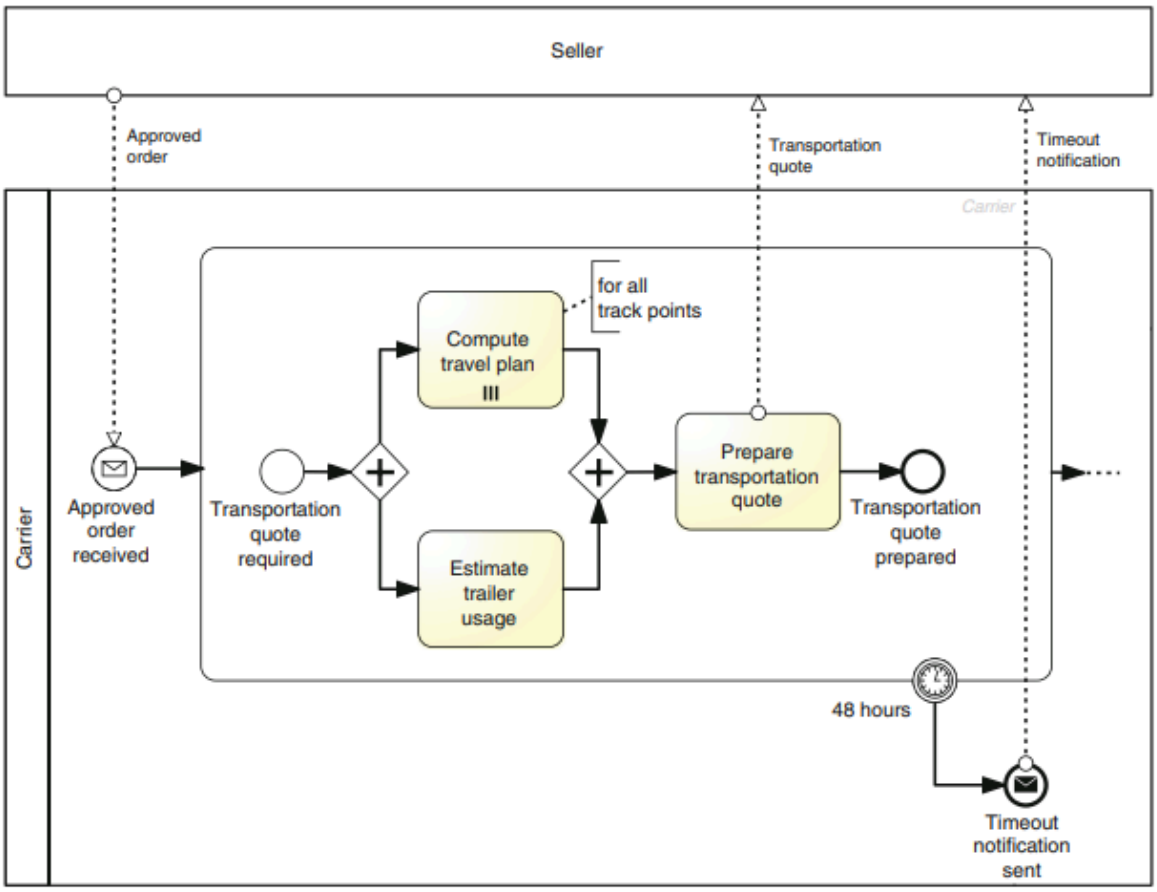
Solution 4.9



Exercise 4.10 Model the following process fragment.

Once a wholesale order has been confirmed, the supplier transmits this order to the carrier for the preparation of the transportation quote. In order to prepare the quote, the carrier needs to compute the route plan (including all track points that need to be traversed during the travel) and estimate the trailer usage (e.g., whether it is a full track load, half track load or a single package). By contract, wholesale orders have to be dispatched within 4 days from the receipt of the order. This implies that transportation quotes have to be prepared within 48 h from the receipt of the order to remain within the terms of the contract

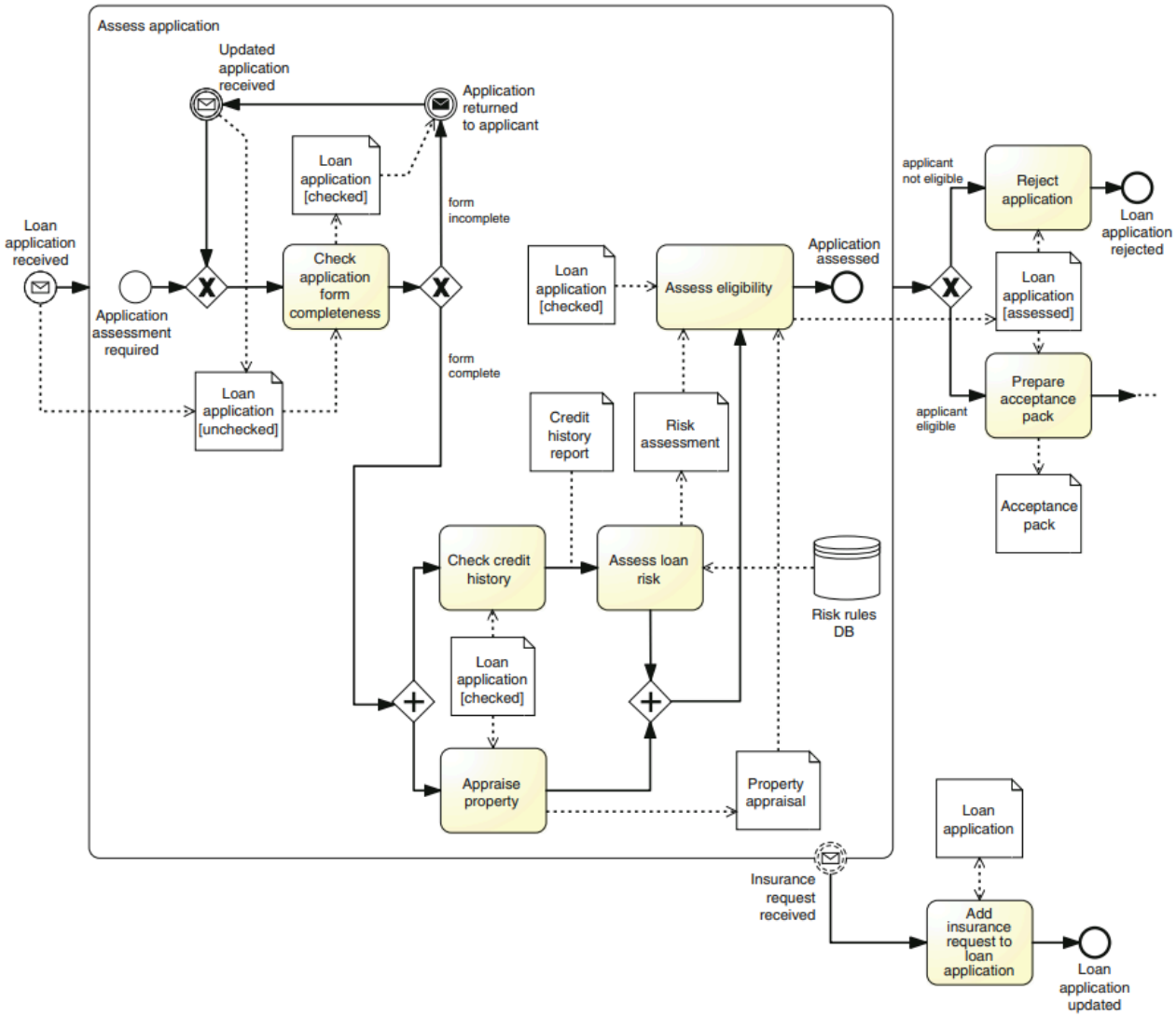
Solution 4.10



Exercise 4.11 Extend the process for assessing loan applications from Solution 3.8 (page 111) as follows.

An applicant who has decided not to combine the loan with a home insurance plan may change its mind any time before the eligibility assessment has been completed. If a request for adding an insurance plan is received during this period, the loan provider will simply update the loan application with this request.

Solution 4.11



Exercise 4.12 Model the following business process for reimbursing expenses.

After an expense report is received from an employee, the employee is notified of the receipt of the report. Next, a new account must be created if the employee does not already have one. The report is then reviewed for automatic approval. Amounts under e 1,000 are automatically approved while amounts equal to or over e 1,000 require manual approval. In case of rejection, the employee must receive a rejection notice by email. In case of approval, the reimbursement is deposited directly to the employee's bank account and an approval notice is sent to the employee via email, with the details of the money transfer. At any time during the review, the employee can send a request for amount rectification. In that case the rectification is registered and the report needs

to be reviewed again. Moreover, if the report is not handled within 30 days, the process is stopped and the employee receives a cancellation notice email so that he can resubmit the expense report from scratch

